

Using HIIT Weapons

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To this point, we've described the history of high-intensity interval training (HIIT) (chapter 1), outlined traditional approaches coaches have used to calibrate their HIIT sessions (chapter 2), explained the general physiological responses (proxy adaptations) associated with HIIT (chapter 3), and broken down HIIT more specifically into the components we can use to form our five key weapons or formats (chapter 4). Recall that these weapons or formats include long intervals (figure 4.13), short intervals (figure 4.14), repeated-sprint training (figure 4.15), sprint interval training (figure 4.16), and game-based HIIT (figure 4.17).

Important components of any armed forces organization throughout history are the concepts of intelligence, tactics, and logistics. In this chapter, we're going to dive much deeper into the intelligence of our weapons so as to understand their intricacies. In doing so, we'll learn to use these weapons to hit the three main physiological targets we're after in our training, specifically the oxidative, anaerobic glycolytic, and neuromuscular responses (chapter 3).

As we've shown in chapter 4, the moving parts of HIIT can be manipulated to form different HIIT weapons or formats, as detailed initially in figure 1.5. These moving parts can be considered useful tools for adjusting HIIT to hit our physiological targets. Most practitioners think only how they'll use a chosen format, be that long intervals, repeated-sprint train-

ing, small-sided games, etc., within a given training program design. But as coaches and practitioners interested in maximizing performance further, we want to extend our thinking to a new level. Instead of thinking only about the format/weapon (what and when), the aim of this chapter is to learn to think more like a four-star practitioner, so as to learn how to fine-tune our high-performance weapon to get the most out of it. That's one of the main aims of this book. Instead of thinking format or weapon first, we'd like to challenge you to think target type first. Based on where we began in chapter 1 (figure 1.5), first we need to understand the sport demands and the context. When ultimately performance is the critical factor, how important is the physical preparation besides the context of the other factors integral to sport performance, such as skill and tactics (figure 1.4)? We need to prioritize our time and resources appropriately. After we're clear on the importance of HIIT, we next need to consider:

1. the sport's or player's physical demands;
2. athlete profiles, including athlete history, training age, and other individual characteristics;
3. long-term adaptation targets; and,
4. training periodization, or how the training plan fluctuates daily, weekly, and season long, depending on our aims and the sport or event schedule.

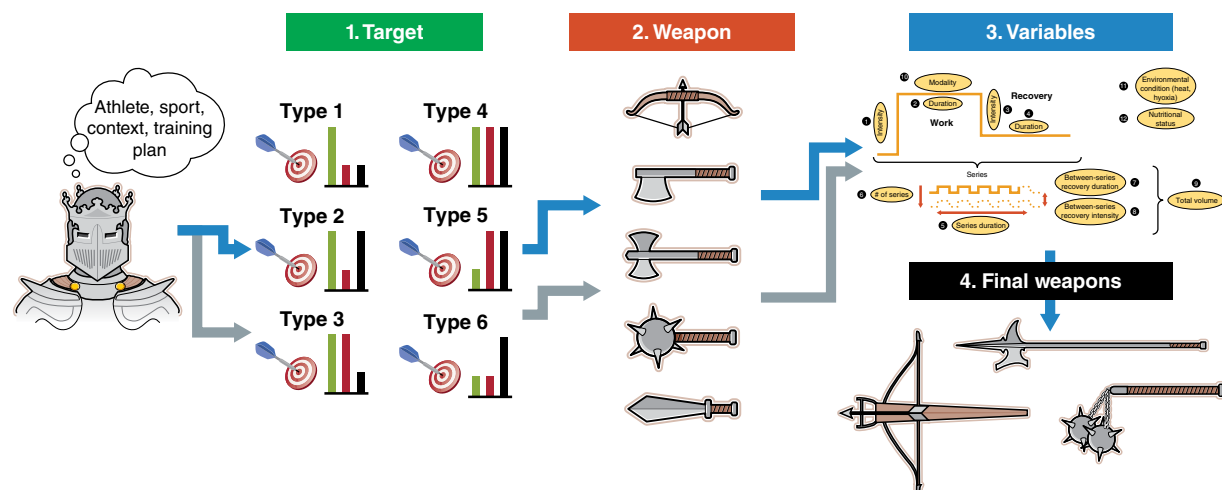


Figure 5.1 The thought patterns that can be used in programming. The more common thought pattern when it comes to HIIT weapon selection is to think format/weapon first. Our challenge to the reader is to think differently: what do you want (HIIT type/target), how do you do it (format/weapon), what variables do you adjust to hit your targets before final creation of your weapon.

As highlighted in the decision tree outlined in figure 1.5, all these factors merge toward establishing the desired physiological objectives of the HIIT session. Are we after aerobic, anaerobic, or neuromuscular responses (proxy adaptation) that we outlined in chapter 3 to develop our athletes appropriately, depending on where things sit in the calendar year relative to our sport performance aims? Again with respect to context, are we clear on the needs related to periodization (e.g., training blocks, within-cycle programming to avoid interference and maximize stimulus; chapter 6)?

Finally, once we're clear on the aim, we can go forward like a four-star practitioner in selecting the appropriate weapon to hit our target (figure 5.1).

We've mentioned the analogy before, but it might be good to review. If we think like a full general given a military assignment by his president, he must consider the targets based on the objectives. He has an arsenal of options at his disposal. He can use a weapon of mass destruction, hitting his target of course, but also everything else in its wake, including civilian casualties or collateral damage. Or, he can use military intelligence, call on his Navy Seal team, bring in Rambo, and be more specific with his approach. While occasionally mass destruction weapons must be used in the context of real-world training application, our preference is to keep collateral damage to a minimum and select a specifically targeted approach (recall toothpaste analogy, chapter 4). Thus, the aim of this chapter

is to enhance our military intelligence by outlining the target types available for each HIIT weapon. This knowledge means we can make better-informed decisions when we select our weapons and adjust these in accordance with how they function to best hit our three main physiological targets: the oxidative, anaerobic, and neuromuscular systems.

By way of emphasis, note that it is not one weapon that achieves a certain HIIT target type but rather multiple weapons that can be used to hit the specific target depending on how they are manipulated. To recap, we suggest there are six key target combinations to aim for to enhance the related systems (outlined in figure 1.5):

- Type 1**, metabolic, eliciting essentially large requirements from the cardiopulmonary system and oxidative muscle fibers;
- Type 2**, metabolic as per type 1, but with a certain degree of neuromuscular strain;
- Type 3**, metabolic as per type 1, but with a large degree of anaerobic glycolytic energy contribution;
- Type 4**, metabolic (oxidative/anaerobic) as per type 3 plus a certain degree of neuromuscular strain;
- Type 5**, essentially a large anaerobic contribution combined with neuromuscular strain; and
- Type 6**, essentially a large neuromuscular strain, with limited metabolic contribution (neither oxidative nor anaerobic).

This chapter describes each weapon, and using our combined experience and history in the academic (121) and applied areas (50, 51), we describe the magnitude of the aerobic oxidative, anaerobic glycolytic, and neuromuscular responses to acute HIIT variable manipulations.

Long Intervals

We lead off this chapter with our first weapon, the long-interval format. Recall that long intervals typically consist of intervals performed near $\dot{V}O_{2\max}$ speed and power, over durations of 2 to 5 min. Long intervals are the weapons we can use to hit target types 3 and 4.

$\dot{V}O_2$ (Oxidative) Responses to Long Intervals

When we think of long intervals, we typically focus on targeting the top end of the aerobic system. As outlined in chapter 3, the hallmark measure of aerobic system function is $\dot{V}O_{2\max}$, considered a proxy of maximal aerobic locomotor function and highly related to endurance exercise performance. Thus, how long (time) we can perform at $\dot{V}O_{2\max}$, referred to as T at $\dot{V}O_{2\max}$ (or $T_{\max}$), becomes a generic measure we can use to observe how the manipulation of different HIIT variables affects the acute oxidative response. While not definitively proven, the principle of specificity suggests that if $\dot{V}O_{2\max}$ is important, then it makes sense to work on strategies that allow the accumulation of T at $\dot{V}O_{2\max}$. This section focuses on the intricacies of manipulating variables within long intervals to enhance $\dot{V}O_{2\max}$. As long intervals are long, such intervals naturally target type 3 and 4 acute oxidative responses (figure 4.13).

Exercise Intensity During Long Intervals

As shown in both chapters 2 and 4, work intensity close to $v/p\dot{V}O_{2\max}$ is needed to elicit a maximal $\dot{V}O_2$ response. When we look to the scientific literature, these responses are typically described during single constant-speed or power exercise, not always applicable to the repeated nature of interval training exercise, but it's a place to start. In an attempt to determine the velocity and exercise intensity associated with the longest T at $\dot{V}O_{2\max}$ during a run to exhaustion, six physical education students performed four separate runs at 90, 100, 120, and 140% of their $v\dot{V}O_{2\max}$ (17 km/h) (23). Not surprisingly, time to exhaustion ($T_{\lim}$) was inversely related to the

running intensity. T at $\dot{V}O_{2\max}$ during 90% and 140% conditions was trivial (i.e., <20 s in average), but reached substantially larger values at 100% and 120%: 190 ± 87 (57% of $T_{\lim}$, ES > +2.8) and 73 ± 29 s (59%, ES > +1.7). In another study, middle-distance runners did not manage to reach $\dot{V}O_{2\max}$ while running at 92% of $v\dot{V}O_{2\max}$ (110). The ability to reach $\dot{V}O_{2\max}$ during a single run at $v\Delta 50$ (i.e., the velocity between the maximal lactate steady state and $v\dot{V}O_{2\max}$, $\approx 92\%$ -93% of $v\dot{V}O_{2\max}$ (23)) via the development of a $\dot{V}O_2$ slow component (5) might be fitness dependent (20), with more highly trained runners less able to reach $\dot{V}O_{2\max}$, possibly due to their higher fat oxidation rates and greater fatigue resistance (105). In addition, as the determination of $v\Delta 50$ is impractical in the field, work intensities of $\geq 95\%$ $v/p\dot{V}O_{2\max}$ are recommended to maximize T at $\dot{V}O_{2\max}$ during single isolated runs.

Of course, in practice, athletes do not exercise to exhaustion, but use intervals or sets. Thus, slightly lower intensities ($\geq 90\%$ $v/p\dot{V}O_{2\max}$) are often used when considering repeated exercise bouts such as long-interval sessions and game-based HIIT, since interval $\dot{V}O_2$ is likely to increase with successive repetitions due to the development of a $\dot{V}O_2$ slow component (5). As suggested by Astrand in the 1960s (5), exercise intensity does not need to be maximal during an HIIT session to elicit $\dot{V}O_{2\max}$.

Time to Reach $\dot{V}O_{2\max}$ and Maximizing Long-Interval Duration

As alluded to in chapter 4, the concept of optimizing long-interval duration with respect to achieving the longest time possible at $\dot{V}O_{2\max}$ was of particular interest to Paul in the early 2000s, and this was one of the primary areas of research investigated during his doctoral studies (121). If $\dot{V}O_{2\max}$ is to be reached during the first interval of a sequence, its interval duration must be at least equal to the time needed to reach $\dot{V}O_{2\max}$. Thus, with short intervals, as during typical HIIT sessions (work interval duration < time needed to reach $\dot{V}O_{2\max}$), $\dot{V}O_{2\max}$ is usually not attained on the first interval. The time needed to reach $\dot{V}O_{2\max}$ during constant-speed/power exercise to exhaustion has received considerable debate in the past (17, 102, 108-110, 123). As shown in figure 5.2, the marker shows considerable variability across a number of factors including fitness, gender, and age, with ranges from 97 s (17) to 299 s (110), along with high inter-subject variability (20%-30% (108, 110, 123) to 40% (102)).

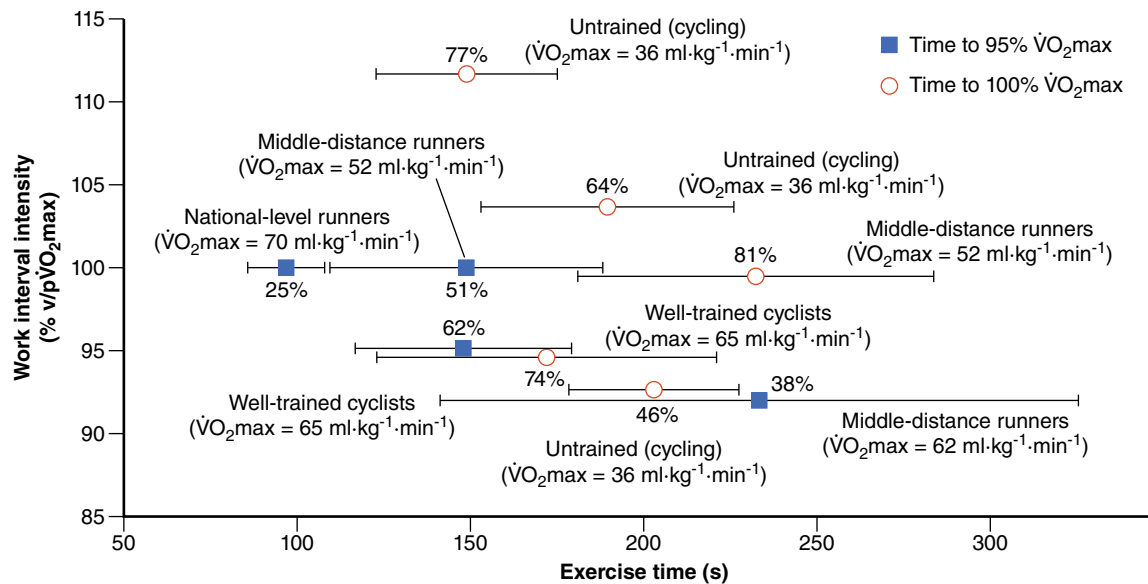


Figure 5.2 Time (SD) to reach 90% or 95% of maximal oxygen uptake ($\dot{V}O_{2\max}$) as a function of exercise intensity in various sports and populations.

While methodological differences likely explain some of these discrepancies (i.e., whether 95% or 100% of $\dot{V}O_{2\max}$ is considered, the presence and type of pretrial warm-up), the variability is consistent with those shown in $\dot{V}O_2$ kinetics at exercise onset. $\dot{V}O_2$ kinetics, that is the time it takes for oxygen delivery to respond to the demands of exercise, are generally affected by exercise intensity (114), accelerated during running compared with cycling (107) and faster in trained individuals (143). The relationship between $\dot{V}O_2$ kinetics at exercise onset and $\dot{V}O_{2\max}$, however, is less clear, with some studies reporting relationships (42) and others showing no correlation (11, 53, 153), suggesting that the $\dot{V}O_2$ kinetics at exercise onset is more related to training status (118, 143) than $\dot{V}O_{2\max}$ per se. For intervals shorter than the time needed to reach $\dot{V}O_{2\max}$, $\dot{V}O_{2\max}$ can still be reached during consecutive intervals, through the priming effect of an adequate warm-up and/or a series of preliminary intervals, which accelerates $\dot{V}O_2$ kinetics (79, 93) and the development of the $\dot{V}O_2$ slow component (5). As illustrated clearly by Burnley et al. (62), a second bout of heavy exercise speeds $\dot{V}O_{2\max}$ attainment by about 2 min (figure 5.3).

As an alternative to the use of fixed long-interval durations, using 50% to 70% of the T_{lim} at v/p $\dot{V}O_{2\max}$ has been suggested as an alternative to individualizing interval training (15, 21, 122, 123, 167, 168). While the rationale of this approach is

sound (50% to 70% is the average proportion of T_{lim} needed to reach $\dot{V}O_{2\max}$), this method lacks practical application in the field. To our knowledge, prescribing training based on time to exhaustion is very rare compared to how endurance athletes actually train for a number of reasons.

1. In addition to v/p $\dot{V}O_{2\max}$, time to exhaustion at v/p $\dot{V}O_{2\max}$ must be determined, which is only a moderately reliable measure (CV = 12% (134) to 25% (22)), is exhaustive by nature and highly depends on the accuracy of the v/p $\dot{V}O_{2\max}$ determination (135).
2. The time required to reach 100% of $\dot{V}O_{2\max}$ has frequently been reported to be longer than 75% of T_{lim} in some participants (108–110).
3. Intervals lasting 70% of T_{lim} have been reported as very difficult to perform, likely due to the high anaerobic energy contribution required (167).
4. For athletes presenting with exceptionally long T_{lim} , repeating sets of 60% of T_{lim} is typically not attainable (123).
5. There is no link between T to $\dot{V}O_{2\max}$ and T_{lim} (108, 123).

In light of these nuances, since a given percentage of the T_{lim} results in vastly different times spent at $\dot{V}O_{2\max}$, the most logical way to use T to $\dot{V}O_{2\max}$ to individualize interval length (108) may be to

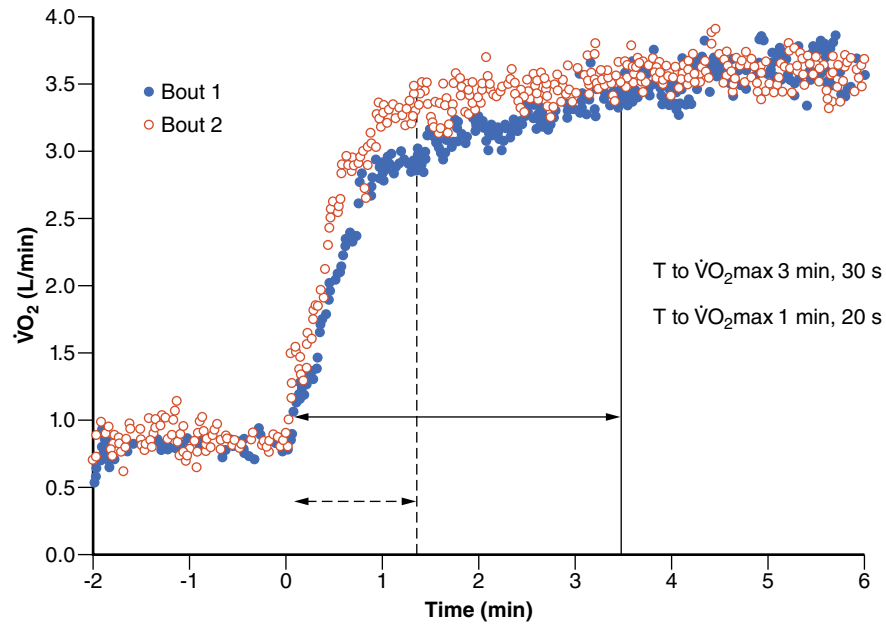


Figure 5.3 The $\dot{V}O_2$ response of a typical subject to a first and second bout of heavy exercise. Note that $\dot{V}O_2$ projects to a higher value after about 2 min in the second bout, indicative of an increased primary $\dot{V}O_2$ amplitude or a faster rate of engagement in the aerobic oxidative energy system.

Adapted by permission from M. Burnley, J.H. Doust, D. Ball, and A.M. Jones, "Effects of Prior Heavy Exercise on $\dot{V}O_2$ Kinetics During Heavy Exercise Are Related to Changes in Muscle Activity," *Journal of Applied Physiology* 93 no. 1 (2002): 167-174.

take the T to $\dot{V}O_{2\max}$ (if known) and add 1 or 2 min. If the T to $\dot{V}O_{2\max}$ cannot be determined (as is often the case), we recommend using fixed long-interval durations of 2 to 5 min. These can be adjusted in accordance with the athlete's exercise mode and training status, with less-trained individuals performing at a lower training intensity but with longer intervals. If we consider that the time constant of the primary phase of the $\dot{V}O_2$ kinetics at exercise onset (τ) in the severe intensity domain is generally in the range of 20 to 35 s (107, 110, 153), and that a steady state ($\geq 95\% \dot{V}O_{2\max}$) is reached after exercise onset within $\approx 4 \tau$, $\dot{V}O_{2\max}$ should theoretically be reached from between 1 min, 20 s to 2 min, 20 s when long intervals are repeated, irrespective of training status and exercise mode. This is consistent with the data shown by Vuorimaa et al. (180) in national-level runners ($v\dot{V}O_{2\max} = 19.1 \pm 1$ km/h), where $\dot{V}O_{2\max}$ values were reached during 2 min intervals but not during 1 min intervals (work:rest ratio 1:1). Similarly, Seiler and Sjursen (163) showed in well-trained runners ($v\dot{V}O_{2\max} = 19.7 \pm 1$ km/h) that peak $\dot{V}O_2$ was only $82\% \pm 5\%$ of $\dot{V}O_{2\max}$ during 1 min intervals, while it reached $92\% \pm 4\%$ during 2 min intervals. Extending the work duration fur-

ther did not modify these peak values ($93\% \pm 5\%$ and $92\% \pm 3\%$ for 4 and 6 min intervals, respectively) (163). Although performed on an inclined treadmill (5%), these latter sessions were completed at sub-maximal self-selected velocities (i.e., 91%, 83%, 76%, and 70% $v\dot{V}O_{2\max}$ for 1, 2, 4, and 6 min intervals, respectively (163)), which probably explains why $\dot{V}O_{2\max}$ was not reached (23).

Volume of HIIT With Long Intervals

Another variable that will contribute to the T at $\dot{V}O_{2\max}$ is the number of long-interval repetitions completed. In reality, few studies have actually examined HIIT sessions or programs consistent with the sessions that athletes actually perform in the field, and research describing T at $\dot{V}O_{2\max}$ in this regard is limited. Billat (15, 21) has often promoted using $2.5 \times T_{\lim}$ for an athlete's individualized target training volume (figure 5.4). Cumulated high-intensity ($>90\% v/p\dot{V}O_{2\max}$) exercise time during typical sessions in well-trained athletes has been reported to be 12 min (6×2 min or $6 \times \approx 600$ m (136)), 15 min (5×3 min or $5 \times \approx 800$ -1000 m (48)), 16 min (4×4 min or $4 \times \approx 1000$ -1250 m (116)), 24 min (6×4 min or $6 \times \approx 1000$ -1250 m (162); 4×6 min or

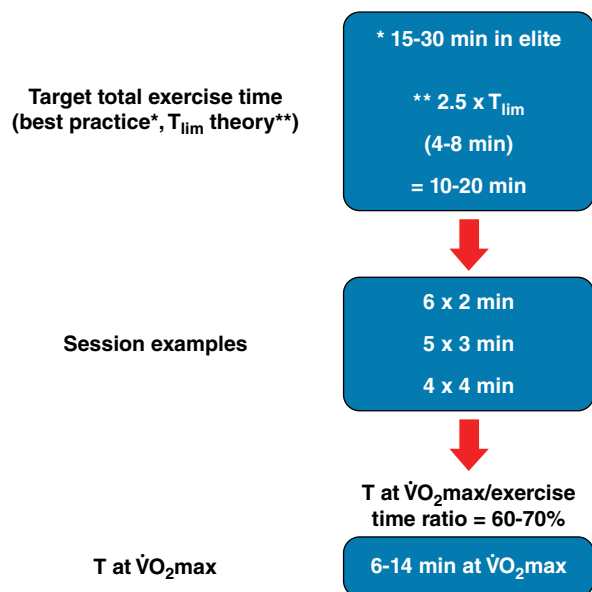


Figure 5.4 Process used to define target time spent at or near $\dot{V}O_{2max}$ (T at $\dot{V}O_{2max}$) during HIIT with long intervals. Data from elite athletes (* best practice) or Billat's recommendations (** T_{lim} theory (21)) suggest that overall exercise time should be between 10 and 30 min. Once total volume is broken into sets, and knowing the average portion of exercise time that is actually spent at or near $\dot{V}O_{2max}$ during each interval (T at $\dot{V}O_{2max}$:exercise time ratio), it is possible to estimate the actual T at $\dot{V}O_{2max}$ associated with those prescribed sessions.

$4 \times \approx 1500$ m (163)) and 30 min (6×5 min or $5 \times \approx 1300$ -1700 m (76)), enabling athletes to accumulate, depending on the HIIT format, from 10 min >90% (48, 136) to 4-10 min >95% (76, 136) at $\dot{V}O_{2max}$. In our experience, elite athletes typically tend to accumulate a greater T at $\dot{V}O_{2max}$ for a given HIIT session at some point of the season. Also of note is that such training can be highly stressful, and inappropriate (excessive) prescription can rapidly lead to signs of overtraining (see chapter 7).

Recovery Interval Characteristics During Long-Interval HIIT

Recovery interval characteristics, both the duration and intensity, were highlighted as one of the 12 manipulation factors of importance in chapter 4. As we discussed, these two variables must be considered in light of maximizing work capacity during subsequent intervals (by increasing blood flow to accelerate muscle metabolic recovery, e.g., phosphocreatine

(PCr) resynthesis, H^+ ion buffering, regulation of inorganic phosphate (Pi) concentration and K^+ transport, muscle lactate oxidation) and maintaining a minimal level of O_2 to speed T to $\dot{V}O_{2max}$ during subsequent intervals (i.e., starting from an elevated baseline) (15, 132). While performing active recovery between interval bouts is appealing to accelerate T to $\dot{V}O_{2max}$, and in turn, induce a higher fractional contribution of aerobic metabolism to total energy turnover (79), its effects on performance capacity (T_{lim} , and hence, T at $\dot{V}O_{2max}$) are not straightforward.

As we determined in chapter 4, and in the context of long-interval HIIT, passive recovery is recommended when the relief interval is less than 2 to 3 min. If an active recovery is chosen for the above-mentioned reasons (15, 79, 132), relief intervals should last at least 3 to 4 min at a submaximal intensity (12) to allow the maintenance of high-intensity exercise during the following interval. Made even simpler, about 2 min of passive walking seems simple best practice for maximizing T at $\dot{V}O_{2max}$ (refer to figure 4.5 and related text for details).

Uphill Running During HIIT With Long Intervals

Recall that hill running was variable 10 (work modality) of the many means we can manipulate to adjust physiological stress during HIIT (chapter 4). Despite its common practice (19), the cardiorespiratory responses to field-based HIIT sessions involving uphill or staircase running have received relatively little attention. Laboratory studies have shown that for a given running speed, $\dot{V}O_2$ is higher during uphill running compared to level running after a couple of minutes, probably due to the increased forces required to move against gravity, the subsequently larger motor units recruited, the greater reliance on concentric contractions, higher step frequency, increased internal mechanical work, shorter swing/aerial phase duration, and greater duty factor (185), all potential initiators of the $\dot{V}O_2$ slow component (152). However, in practice, athletes generally run slower on hills versus the track (165). Gajer et al. (90) found in elite French middle-distance runners ($v\dot{V}O_{2max} = 21.2 \pm 0.6$ km/h, $\dot{V}O_{2max} = 78 \pm 4$ mL $\cdot$ min $^{-1}$ ·kg $^{-1}$) that T at $\dot{V}O_{2max}$ observed during a hill HIIT session (6×500 m (1:40), 4 to 5% slope [$85\% v\dot{V}O_{2max}$]/1:40 [0%]) was lower compared to a reference track session (6×600 m (1:40) [102% $v\dot{V}O_{2max}$]/1:40 [0%]). While $\dot{V}O_2$ reached 99% and 105% $\dot{V}O_{2max}$ during the hill and track sessions,

respectively, the T at $\dot{V}O_{2\max}$:exercise time ratio was *moderately* lower during the hill HIIT (27% versus 44%, ES ≈ -1.0). The reason for the lower T at $\dot{V}O_{2\max}$ during the hill HIIT is unclear. Despite the expected higher muscle force requirement during hill running (165), this is unlikely enough to compensate for the reduction in absolute running speed. If we consider that running uphill at 85% $\dot{V}O_{2\max}$ with a grade of 5% has likely the same (theoretical) energy requirement as level running (or treadmill running with a 1% grade to compensate for wind resistance (154)) at $\approx 105\%$ (138), the differences observed by Gajer et al. could have been even greater if the flat condition were run at a faster (and possibly better matched) speed (105% versus 102% (90)). As well, intervals in these sessions might not have been long enough to observe the additional slow component generally witnessed with uphill running (≈ 2 min (152)).

More recently, Barnes et al. (10) used the same approach to examine multiple variable manipulations (slopes) as the study by Stepto and colleagues (169) described in chapter 4. Here the authors used five different hill repeat-interval programs (2 times/wk) in small groups (3 to 5 runners) over 6 wk. Intervals were performed using gradients ranging from 4% to 18%, with associated longer or shorter bout durations (faster or slower run speeds) accordingly. Like Stepto et al. (169), no clear winner was observed from the formats in 5 km time trial performance. Interestingly, however, in the two middle groups performing intervals in the long-interval domain (G3 and G4) at 100% and 90% of $\dot{V}O_{2\max}$, with interval repetitions of ~ 2.5 and ~ 5 min, respectively, the trend was toward improving $\dot{V}O_{2\max}$ the most after 6 wk. However, these findings likely relate simply to the specificity of the HIIT programs used. (The specific long-interval format improved $\dot{V}O_{2\max}$ the most as it often does.) From a practical and applied standpoint, training programs are not ever exclusive to a single format, and an athlete's brain (central governor) will always lower and raise intensity in accordance with what is manageable and appropriate. Nevertheless, with respect to long intervals and hill gradient, research would suggest that a 7% to 10% gradient is an appropriate slope for performing hill-based long intervals in well-trained runners.

Summary of Oxidative Responses to Long Intervals

The long interval is a key weapon in our arsenal that we can use to enhance the oxidative response to

HIIT. To achieve the necessary oxidative response (proxy adaptation), we have learned that for long intervals

- intensity should range from 92% to 102% of $p/\dot{V}O_{2\max}$,
- duration should span 2 to 5 min, at a repetition number that achieves ~ 10 min of T at $\dot{V}O_{2\max}$,
- recovery intervals should be passive (i.e., walking) for about 2 min, and
- in the case of runners using hills, a 7% to 10% gradient appears appropriate.

However, despite this ideal HIIT profile, practitioners use many format variations in their programs (i.e., longer intervals, different work:rest ratio, etc.). The decision to use a variation of this weapon over another depends on numerous factors, including an athlete's needs, the sport's needs, and context. Figure 5.5 shows how four different HIIT sequences, which all possess a similar oxidative target by allowing an athlete to spend 8 to 10 min at $\dot{V}O_{2\max}$, can vary substantially in terms of training volume, anaerobic contribution, and musculoskeletal and neuromuscular strain (total distance and distance at high speed). There is no right or wrong option, but practitioners may choose between options to either increase training volume (2 sessions in the middle of figure 5.5, >35 min) or limit blood lactate accumulation (session on the left) and high-speed running volume (SSGs, on the right). Note in this example the different responses in populations examined (runners for the three sessions on the left versus handball players for the SSGs). This explains the high lactates after the SSGs for the handball players, who would probably not cope with the running sessions applied to the runners (figure 5.5).

The following sections will extend our reasoning for the manipulation of the different HIIT variables to better control anaerobic contribution and musculoskeletal or neuromuscular strain during long intervals.

Anaerobic Glycolytic Energy Contribution to Long-Bout-Duration HIIT Sessions

Though not typically considered relative to the larger aerobic contribution to long intervals, the anaerobic glycolytic energy contribution to long intervals can be substantial. For example, in endurance-trained athletes performing constant-speed efforts at

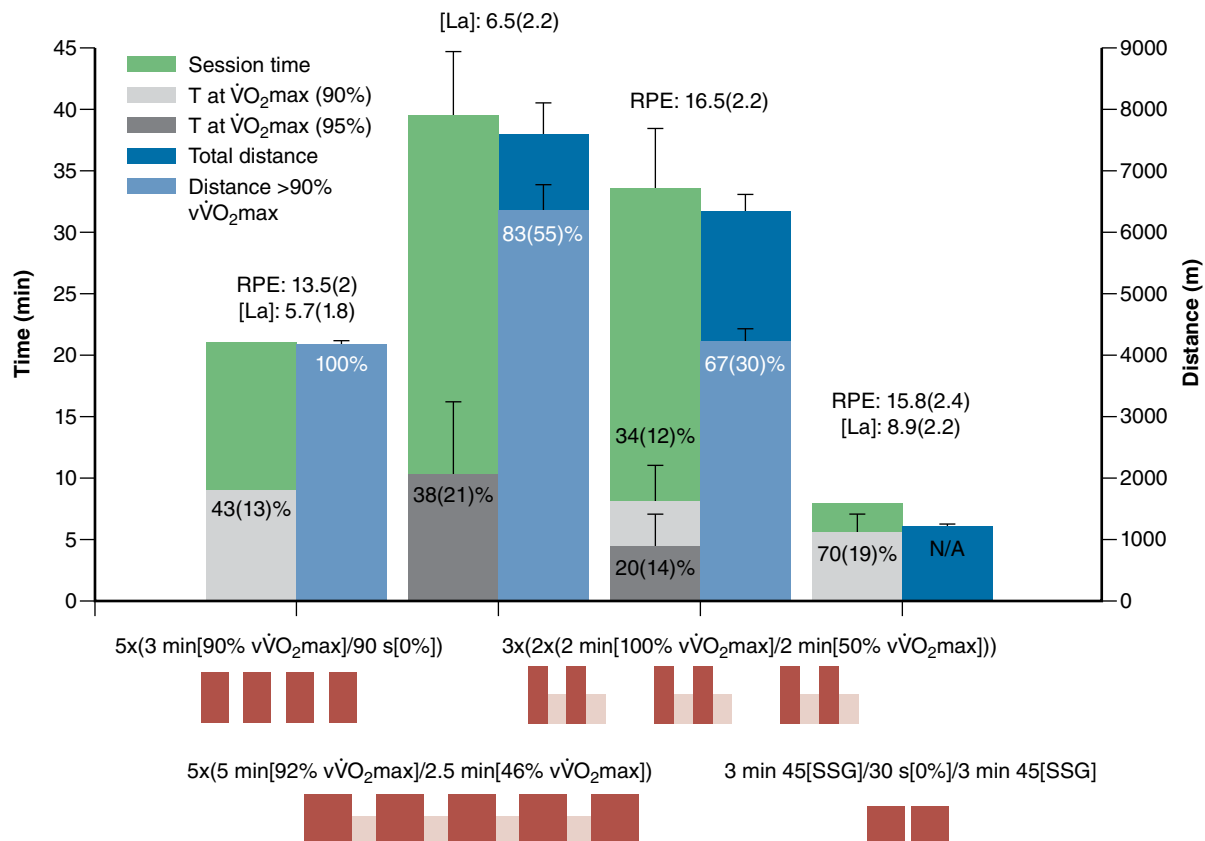


Figure 5.5 $\dot{V}O_2$ responses of four distinct long-interval HIIT sessions, highlighting the impact of different variable manipulations on T at $\dot{V}O_{2max}$, distance run, and session time (48, 56, 76, 136). Mean $\pm$ SD of total session time, T at $\dot{V}O_{2max}$, total distance, and distance run above 90% of $v\dot{V}O_{2max}$ during four different long-interval HIIT sessions. Percentages (mean $\pm$ SD) refer to T at $\dot{V}O_{2max}$ relative to the total session time and distance run above 90% of $v\dot{V}O_{2max}$ relative to the total distance run. RPE and [La] (mmol/L) are provided as mean $\pm$ SD when available. HIIT: high-intensity interval training; [La]: blood lactate concentration; N/A: not available; RPE: rating of perceived exertion; SSGs: small-sided games (handball); $\dot{V}O_{2max}$: maximal oxygen uptake; T at $\dot{V}O_{2max}$: time spent above 90% or 95% of $\dot{V}O_{2max}$; $v\dot{V}O_{2max}$: minimal running speed associated with $\dot{V}O_{2max}$.

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$v\dot{V}O_{2max}$, over interval durations longer than ~90 s, the initial rise in blood lactate ranges from 5 to 7 mmol·L⁻¹·5min⁻¹ (figure 5.6) (17, 76, 133, 162, 167, 170). Thus, long intervals can be manipulated to target the anaerobic glycolytic energy component associated with type 3 and 4 acute responses.

Effect of Work Interval Intensity on the Anaerobic Glycolytic Response to Long Intervals

While manipulation of HIIT variables can modulate the anaerobic glycolytic response, it must be appreciated that blood lactate accumulation is additionally related to training status (14) (figure 5.6). For

example, in elite French middle-distance runners ($v\dot{V}O_{2max}$ = 21.2 $\pm$ 0.6 km/h) performing repeated 600 m bouts (~1:40, work:relief ratio = 1), the initial rate of blood lactate increase was in the lower range of the values reported, i.e., ~5 mmol·L⁻¹·5min⁻¹ (90). The shorter work duration in this example (100 s) also illustrates the important concept that shorter interval repetitions will lower the lactate response (see subsequent section).

Despite the lack of a direct examination, study comparisons suggest that higher work intensities performed during long intervals likely elicit increased rates of blood lactate production (proxy anaerobic glycolytic energy contribution) (162, 167, 170).

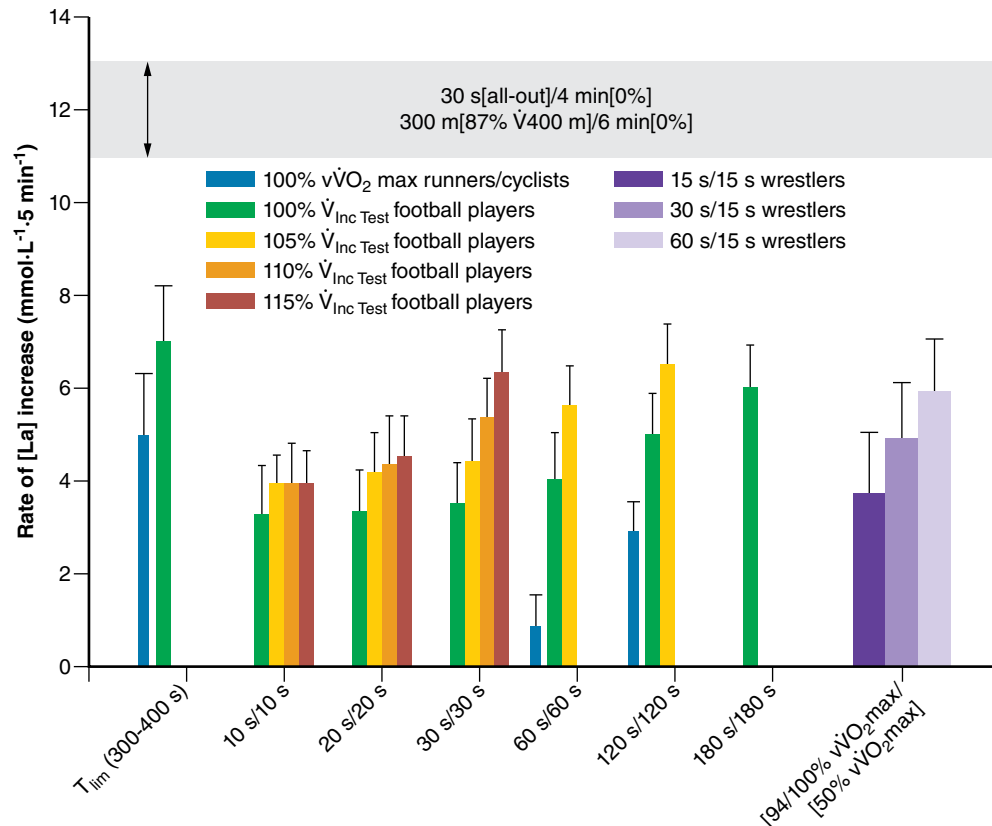


Figure 5.6 Estimated anaerobic energy contribution to different short- and long-HIIT formats as inferred from the initial rate of blood lactate accumulation ([La]). Values are mean $\pm$ SD. Exercise intensities are % of the peak speed reached in the 30-15 intermittent fitness test (V_{IFT}), speed or power associated with maximal oxygen uptake ($v/\dot{V}O_2\max$), or peak incremental test speed ($V_{Inc,Test}$). Gray rectangular box represents the initial rate of blood lactate accumulation during lactate production training, i.e., all-out sprints or at the speed maintained during a 400 m run (V_{400m}).

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Effect of Work Interval Duration on the Anaerobic Glycolytic Response to Long Intervals

Extending the interval duration of a long-interval HIIT session without altering the relief interval duration clearly increases the anaerobic glycolytic energy contribution, as more work must be completed in a given time period. In practice, however, coaches generally maintain the work:relief ratio when they manipulate their HIIT variables. In these latter conditions, an increase in work interval duration additionally is likely to increase anaerobic glycolytic energy contribution. For instance, doubling interval duration (2 min versus 1 min run at $\dot{V}O_2\max$, work:relief ratio = 1) leads to substantial

increases in anaerobic glycolytic energy contribution (figure 5.6) (180). Accumulated O_2 deficit ($\approx 25 \pm 2$ versus 21 ± 2 mL/kg, $ES = +2.3$) and end-session blood lactate measurements (8.8 ± 3.6 versus 4.8 ± 1.1 mmol/L, $ES = +1.7$) were *very largely* and *largely* higher, respectively. Likewise, in distance runners ($\dot{V}O_2\max = 19.5 \pm 0.7$ km/h), increasing the interval duration run at $\dot{V}O_2\max$ from 2:10 to 2:30 (+15%) with a work:relief ratio of 1:2 resulted in an almost twofold increase in the initial rate of blood lactate accumulation (from ≈ 6 to ≈ 10 mmol·L $^{-1}$ ·5min $^{-1}$) (167). Of note, however, is that a decrease in work intensity (i.e., from 93% to 84% of $\dot{V}O_2\max$ on a treadmill) can compensate for the effect of work interval extension from 1 to

6 min (work:relief ratio = 1) and maintains blood lactate at manageable levels (i.e., 4-5 mmol/L post exercise) (163).

Effect of Recovery Interval Characteristics on the Anaerobic Glycolytic Response to Long Intervals

As we stated with this factor in chapter 4, the influence of HIIT recovery interval duration on the subsequent bout's anaerobic glycolytic energy contribution is not straightforward. Maintaining a work:relief ratio of 1, accumulated O_2 deficit in the first min of a 2 min/2 min sequence run at $\dot{V}O_{2max}$ has been shown to be markedly greater than a 1 min/1 min set (23.8 ± 1.6 versus 20.5 ± 1.9 mL O_2 /kg, ES = 1.9) (180). This is likely related to the lower total $\dot{V}O_2$ attained in the longer 2 min rest period condition, which increases the O_2 deficit for subsequent intervals (160). When the duration of the exercise interval is fixed, things are a bit different, and a shortening of the recovery interval would be typically associated with higher anaerobic glycolytic energy contribution due to the increased exercise load (greater total work done in less time). However, following self-selected 4 min work interval intensities ranging from 83% to 85% $\dot{V}O_{2max}$ (with a 5% grade), reducing the relief interval from 4 to 1 min did not affect end-session blood lactate levels (6-7 mmol/L) in moderately trained runners ($\dot{V}O_{2max} = 17.6 \pm 1$ km/h) (162). It is worth noting, however, that despite the inclined treadmill, the exercise intensity was submaximal in these studies (162, 163). It is expected that for sessions run at $\dot{V}O_{2max}$, blood lactate responses would be higher and more responsive to relief interval manipulation (167). It is also worth noting that blood lactate concentration, as a systemic measure with a certain inertia, is likely a less sensitive measure of anaerobic glycolytic energy contribution than accumulated O_2 deficit, which may explain the differences between the studies by Vuorimaa et al. (180) and Seiler and Hetlelid (162).

When the relief interval duration is >3 to 4 min, active recovery (60%-70% $\dot{V}O_{2max}$) can be used to accelerate blood lactate clearance compared to passive conditions (3, 12), leading to a lower accumulation throughout the session. While the effect of short (≤ 2 min) passive versus long (>3-4 min) active recovery on the anaerobic energy contribution to exercise has not been studied, it is worth noting that reductions in blood lactate may be indicative of effective lactate shuttling and high

lactate consumption in working skeletal muscles (31). As we touched on in chapter 4, blood lactate gets a bad name in the press, when at the end of the day it's simply a metabolite in flux. Its so-called removal, or lack thereof, in most contexts, is typically irrelevant.

Work Interval Modality Effect on the Anaerobic Glycolytic Response to Long Intervals

While field-based HIIT sessions involving hill repeats, sand running, stair climbing, and plyometric work are very common in the applied setting, there is surprisingly limited data showing the anaerobic energy contribution of such exercise. In elite French middle-distance runners ($\dot{V}O_{2max} = 21.2 \pm 0.6$ km/h, $\dot{V}O_{2max} = 78 \pm 4$ mL $\cdot$ min $^{-1}$ ·kg $^{-1}$), blood lactate accumulation during a self-paced HIIT hill session on the road (6 $\times$ 500 m [~ 1 min, 40 s, work:relief ratio = 1], slope: 4%-5%) was *largely* lower compared with a reference self-paced track session (6 $\times$ 600 m) ([La] post-series: 8.5 ± 2.2 vs. 13.2 ± 4 mmol/L, ES = -1.5) (90). Nevertheless, this could have been due to the lower absolute running speed attained during the inclined condition used in this study that did not compensate for an eventual higher mechanical muscle demand and/or change in muscle pattern activation (165). Further studies are needed to compare the effect of different exercise modes (e.g., running versus cycling versus rowing) on anaerobic glycolytic energy contribution during actual HIIT sessions in athletes. In general, however, for coaches and athletes in the applied setting, exercise mode has a much lower impact on anaerobic glycolytic impact relative to key aspects such as intensity and duration of the work bout.

In summary, while HIIT with long intervals is likely the best format for adapting cardiopulmonary oxidative function, blood lactate accumulation (and likely anaerobic glycolytic energy contribution) can still reach high levels (rate of accumulation >5 mmol $\cdot$ L $^{-1}$ ·min $^{-1}$ and end-session values >10 mmol/L) (167) (figure 5.6), type 3 and 4. Therefore, for coaches striving to limit the anaerobic glycolytic contribution, the use of different forms of HIIT, such as short intervals, may be warranted (15).

Neuromuscular Responses to Long Intervals

The importance of quantifying the effect that an HIIT bout has on the neuromuscular system was highlighted in chapter 3. Long intervals can be used to

hit type 4 targets, which represent an important musculoskeletal and neuromuscular strain. With respect to long intervals, manipulating interval duration on changes in CMJ height were examined in national level runners ($\dot{V}O_{2\max}=19.1 \pm 1$ km/h) by Vuorimaa et al. (179), who showed no within- or between-HIIT differences in CMJ height with 1 versus 2 min interval sessions (1:1 work:rest ratios) ran at $\dot{V}O_{2\max}$ (179). Similarly, there was no effect on stride length throughout successive intervals (179). These findings may relate to the endurance-trained athlete's particular neuromuscular profile, the fact that running speed was similar between the 2 HIIT sessions, and that blood lactate levels remained moderate, even during the 2 min intervals (8.8 ± 3.6 mmol/L). Typically, neuromuscular impairment of muscle function during high-intensity exercise is accompanied by high blood lactate levels, i.e., >10 - 12 mmol/L (99, 144).

The effect of running speed or intensity on neuromuscular load can be indirectly inferred from results reported in highly trained young runners ($\dot{V}O_{2\max}=18.6 \pm 0.9$ km/h) (48). Here, Buchheit et al. (48) examined CMJ height and 5 m sprint times before and after long intervals (5×3 min) in normoxia (90% $\dot{V}O_{2\max}$) and normobaric hypoxia (inspired O_2 fraction = 15.4%, simulated altitude of 2400 m, 84% $\dot{V}O_{2\max}$). For both HIIT protocols, no changes in CMJ height were found, either immediately or 20 min after. Interestingly, while there was no change in sprint times following the runs completed at 84% $\dot{V}O_{2\max}$, sprint times were improved after the runs completed at 90% $\dot{V}O_{2\max}$ immediately after the session (ES for difference in changes = 0.2-0.5), suggestive of a postactivation potentiation effect (figure 5.7) (113). However, this response was no longer evident 4 h later.

Last, compared to a reference track session (6×600 m), stride frequency (2.98 versus 3.1 strides/s) and amplitude (185 versus 203 cm) tended to be lower during a road-based HIIT hill session (6×500 , 4%-5% incline) (90), suggesting that hill training can be effective at lowering hamstring muscle loading. In alignment with this, inclined (5%) 250 m sprints were associated with reduced stride length (-14%) and rate (-7%) and a +27% increase in push-off time, compared with 300 m sprints on the track (165). Importantly, despite no adjustment in quadriceps muscle activation, hamstring muscles were also less activated. Finally, in middle-distance runners ($\dot{V}O_{2\max}=21.8 \pm 1.8$ km/h), performance measures reflecting muscle power have also been

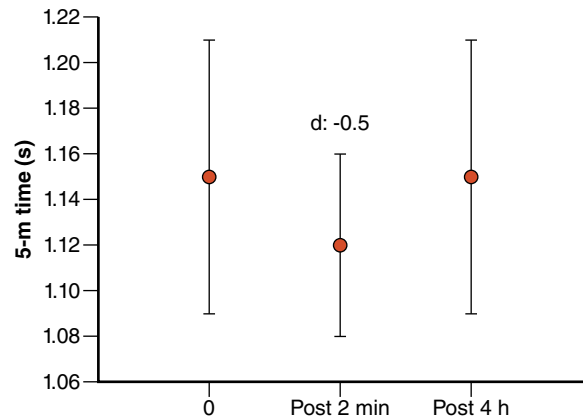


Figure 5.7 Changes in 5 m split time (SD) after long interval HIIT run at 90% $\dot{V}O_{2\max}$. Changes in 5 m split time were improved immediately (post 2 min) after the HIIT session run at 90% $\dot{V}O_{2\max}$, however, there was no substantial change in sprint time following the HIIT session run at 84% in hypoxia (i.e., consistently 1.15 ± 0.06 s).

shown to be lower for uphill versus horizontal incremental running performance (149). Taken together, these data (90, 149, 165) and others (100) suggest that incline running lowers hamstring strain in runners, which could be beneficial to prevent injuries during maximal and/or high-volume sessions (178). However, since athletes might be required to run downhill to prepare for the next interval, care should be taken with respect to potential acute muscle damage arising from the downhill phase (63).

Thus, HIIT hill sessions represent a useful alternative for reducing overall acute hamstring strain and appear well suited for high-volume training cycles, but strategies to avoid the downhill running phase should be considered by coaches. It is also worth noting that over longer time frames (once acute muscle damage is recovered), downhill running can be considered advantageous as it has a prophylactic effect (63) (i.e., repeated-bout effect) and can additionally be effective training for preventing future musculoskeletal injuries (see chapter 14; road running).

The level of neuromuscular strain during long intervals for cyclists can be manipulated by varying cadence or pedaling rate and associated resistance per pedal stroke. For example, Nimmerichter et al. (142) compared the effects of a low- (60 rpm uphill) or high-cadence (100 rpm level ground) long-HIIT (6×5 min) in well-trained cyclists over 4 wk. The 60 rpm group increased power output during both up ($4.4\% \pm 5.3\%$) and flat ($1.5\% \pm 4.5\%$) time trials,

while changes in the level-ground HIIT group were less impressive ($-1.3\% \pm 3.6\%$ and $2.6\% \pm 6.0\%$, for uphill and level ground, respectively). Thus, increased neuromuscular demands (higher forces per pedal stroke) during long-interval HIIT in cyclists may also be beneficial to cycling performance, as alluded to further in our specific application chapter on this sport (see chapter 15, Road Cycling).

Short Intervals

The next weapon we'll describe in detail is the short interval. This is probably the most useful and versatile format for practitioners due to the fact that we can hit almost all HIIT target types and explains why the weapon has received so much attention in the research literature. As alluded to in chapter 4, Martin is particularly passionate about this weapon based on his history, beginning in his youth and continuing through to his coaching and research work. In this section, we will show how short intervals, with bout durations ranging from 10 to 60 s, can be used to target type 1, 2, 3, and 4 acute responses.

O₂ (Oxidative) Responses to Short Intervals

The marker of acute aerobic response, T at $\dot{V}O_{2\max}$, during short-interval runs to exhaustion has largely been shown to be related to the total exercise time (i.e., $T_{\lim}$) (132). This intuitively implies that the first approach you can make toward maximizing T at $\dot{V}O_{2\max}$ during such sessions should be to focus on how you can adjust aspects of the interval format (intensity and duration of work and relief bouts) to increase the $T_{\lim}$. In practice, of course, coaches will almost never prescribe HIIT sessions to exhaustion; they prescribe a series or set of short intervals (56, 136, 137, 173). In this context, it's important to consider the strategies needed to maximize T at $\dot{V}O_{2\max}$ within a given time period or to define time-efficient HIIT formats with respect to the T at $\dot{V}O_{2\max}$ /exercise time ratio. That is, we need to determine the greatest T at $\dot{V}O_{2\max}$ that can be achieved within the total duration of the HIIT session we have available, not including the warm-up. In team sports especially, conditioning coaches often have only 10 to 20 min per session to develop the aerobic qualities of their players (see team sport chapters), so they need to ensure this time is used wisely.

Let's explore the variables we can manipulate to achieve this objective.

Exercise Intensity During Short Intervals

Billat et al. (18) were the first to show the effect of exercise intensity on T at $\dot{V}O_2$ during HIIT using 15 s/15 s short intervals in a group of senior (average age 52 yr) distance runners ($\dot{v}\dot{V}O_{2\max} = 15.9 \pm 1.8$ km/h). While the concurrent manipulation of the relief interval intensity (60%-80% of $\dot{v}\dot{V}O_{2\max}$, which maintained an average HIIT intensity of 85%) might have partially influenced the $\dot{V}O_2$ responses, the authors showed that increasing work interval intensity from 90% to 100% of $\dot{v}\dot{V}O_{2\max}$ was associated with a *small* improvement in the T at $\dot{V}O_2$ /exercise time ratio (81% versus 68%, $ES = +0.5$). However, the T at $\dot{V}O_2$ /exercise time ratio (85%) was not substantially greater when the work interval was increased to 110% compared to 100% of $\dot{v}\dot{V}O_{2\max}$ ($ES = +0.2$). The differences may be related to the fact that the lower amplitude (difference between exercise and recovery intensities) likely accelerates the T to $\dot{V}O_{2\max}$ (see figure 5.8).

Using a fixed relief interval intensity (figure 5.9) (83, 137, 175), increasing work intensity from 100% to 110% of $\dot{v}\dot{V}O_{2\max}$ during a 30 s/30 s format in trained young runners ($\dot{v}\dot{V}O_{2\max} = 17.7 \pm 0.9$ km/h) induced a *moderate* increase in the T at $\dot{V}O_2$ /exercise time ratio ($ES = +0.6$), despite *very large* and *moderate* reductions in $T_{\lim}$ ($ES = -4.4$) and T at $\dot{V}O_{2\max}$ ($ES = -0.7$), respectively (175). A slight increase in work intensity from 100% to 105% of $\dot{v}\dot{V}O_{2\max}$ during a 30 s/30 s short-interval format in well-trained triathletes ($\dot{v}\dot{V}O_{2\max} = 19.8 \pm 0.93$ km/h) was associated with a *large* improvement in the T at $\dot{V}O_2$ /exercise time ratio ($ES = +1.2$) (137). The twofold magnitude difference in the Millet et al. (137) study compared to Thevenet et al. (175) ($ES: +1.2$ versus $+0.6$) is likely due to the fact that Millet's runs were not performed to exhaustion but implemented with predetermined sets. It is therefore possible that if the runs at 100% had been performed to exhaustion (137), this would have compensated for the lower efficiency of the protocol and decreased the difference in T at $\dot{V}O_{2\max}$ observed. Similarly, increasing the work intensity from 110% to 120% of $\dot{v}\dot{V}O_{2\max}$ during a 15 s/15 s format in physical education students ($\dot{v}\dot{V}O_{2\max} = 16.7 \pm 1.3$ km/h) led to a *large* improvement in the T at $\dot{V}O_2$ /exercise time ratio ($ES = +1.8$) (83). Interestingly, in the study by Millet et al. (137), individual improvements in T at $\dot{V}O_2$ with the increase in work intensity were inversely correlated with the athletes' time constant for $\dot{V}O_2$ kinetics at exercise onset ($r = 0.91$, 90% CL (0.61; 0.98)), sug-

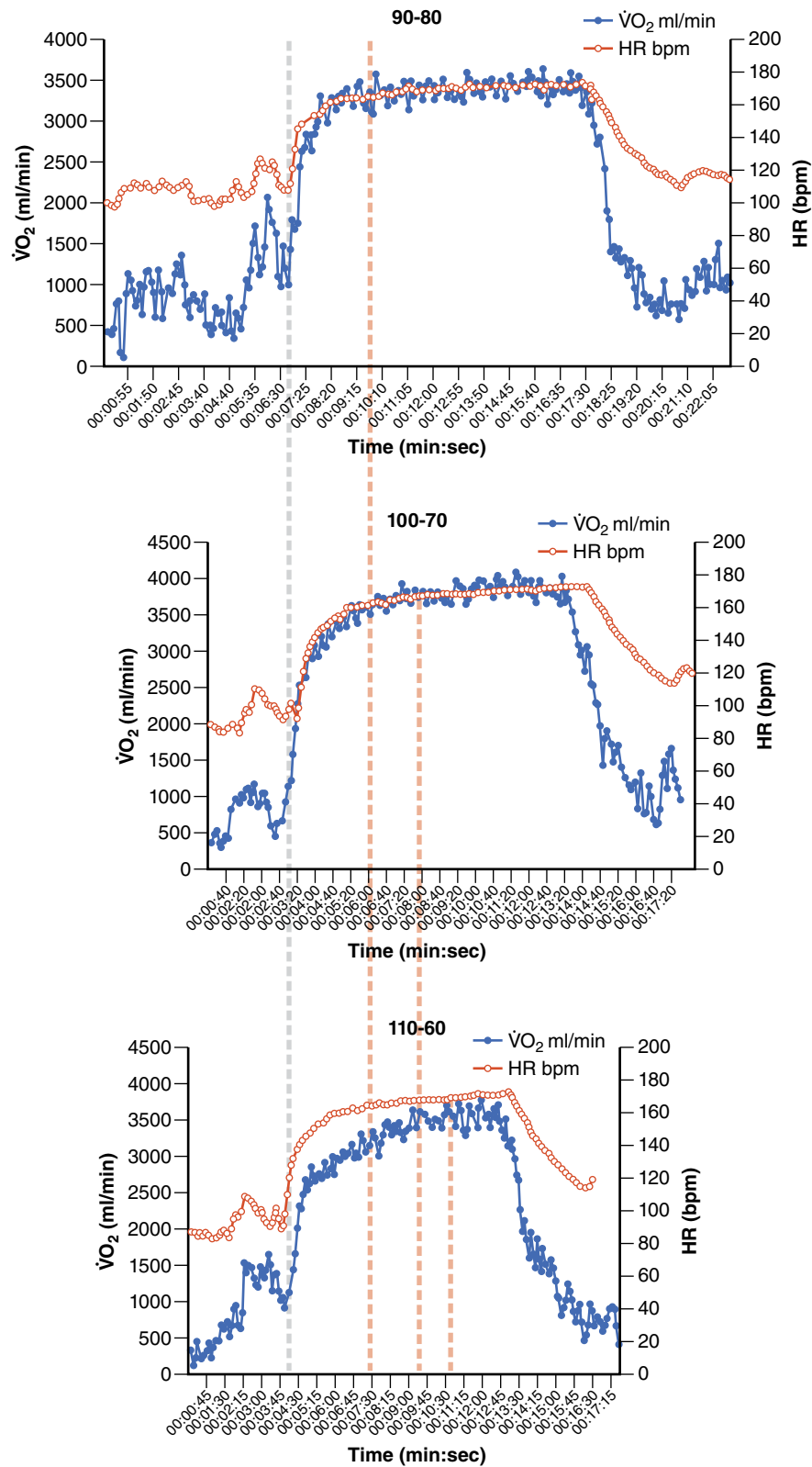


Figure 5.8 Lower amplitude (difference between exercise and recovery intensities) accelerates the T to $\dot{V}O_{2\max}$ (red vertical dotted lines), as shown in the study by Billat et al. (18) using 15 s/15 s short intervals with work:relief intensities of 90%:80%, 100%:70%, and 110%:60% of $\dot{V}O_{2\max}$.

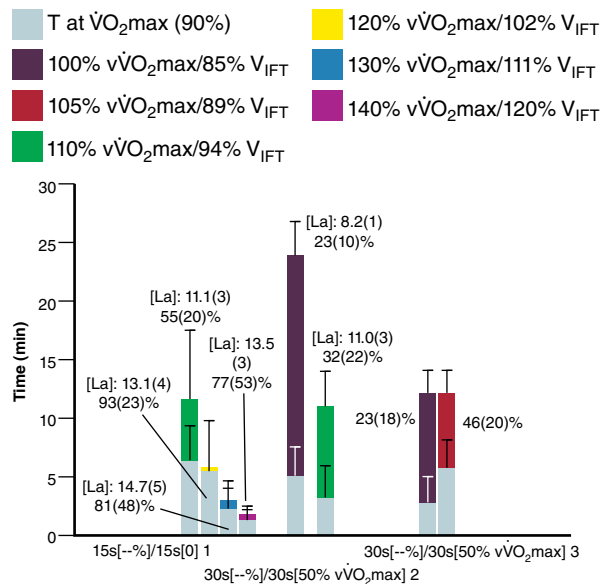


Figure 5.9 Mean $\pm$ SD total time and T at $\dot{V}O_{2max}$ during different short-interval HIIT as a function of changes in work interval intensity. Percentages refer to the mean $\pm$ SD T at $\dot{V}O_{2max}$ relative to total session time (83, 137, 175). [La]: post-exercise blood lactate.

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gesting that the time constant could be an important variable to consider when selecting HIIT variables (35, 137). Practically speaking, this data could imply that coaches should program short intervals at slightly higher intensities for athletes presenting with slow $\dot{V}O_2$ kinetics (i.e., older/less trained (143)), or for athletes exercising on a bike (107). However, since increasing exercise intensity has other implications (e.g., greater anaerobic energy contribution, higher neuromuscular strain), such programming manipulations need to take a cost/benefit approach.

With respect to the use of very high exercise intensities ($>102\%/120\% V_{IFT}/v\dot{V}O_{2max}$) for HIIT, while the T at $\dot{V}O_2$ /exercise time ratio is high (81 and 77% at 130 and 140% of $v\dot{V}O_{2max}$, respectively), exercise capacity is typically impaired and hence, total T at $\dot{V}O_2$ for a given HIIT session is usually relatively low (83) (i.e., 5 min, 47 s at 120% $v\dot{V}O_{2max}$ (83)). Nevertheless, the use of repeated series of such training can allow the accumulation of a sufficient T at $\dot{V}O_{2max}$. Additionally, well-trained athletes are generally able to perform HIIT at this intensity for longer periods (i.e., >8 min (54, 56,

60, 80)), especially when V_{IFT} , instead of $v\dot{V}O_{2max}$, is used (34).

Effect of Work Interval Duration on T at $v\dot{V}O_{2max}$

The effect of work interval duration on systemic $\dot{V}O_2$ responses during short-interval HIIT was one of the first parameters examined in the HIIT literature (6, 68). Surprisingly, however, there is little data available on repeated efforts lasting less than 15 s (figure 5.10), despite the common approach used by coaches (e.g., 10 s/10 s, 10 s/20 s) (26, 74). During very short runs (<10 s), energy requirements in working muscle are met predominantly by oxidative phosphorylation, with more than 50% of the $\dot{V}O_2$ used derived from oxymyoglobin stores (6). During the recovery periods, oxymyoglobin stores are rapidly restored and then available for the following interval (6). As a result, the cardiopulmonary responses of such efforts are relatively low (13), unless exercise intensity is set at a very high level and/or relief intervals are short and intense enough so that they limit complete myoglobin resaturation. Therefore, in the context of HIIT involving short intervals (100%-120% $v\dot{V}O_{2max}$ or 89/105% V_{IFT}), work intervals ≥ 10 s may be required to elicit high

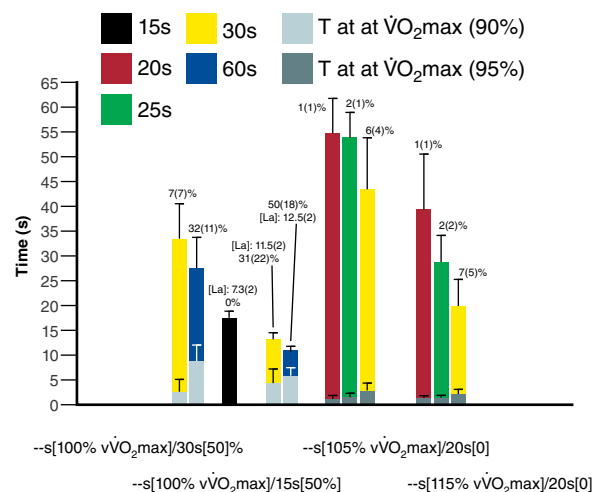


Figure 5.10 Mean $\pm$ SD total time and T at $\dot{V}O_{2max}$ during different short-interval HIIT as a function of changes in work interval duration. Percentages refer to the mean $\pm$ SD T at $\dot{V}O_{2max}$ relative to total session time. [La]: post-exercise blood lactate.

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$\dot{V}O_2$ responses. Indeed, prolonging exercise duration increases the relative aerobic energy requirements (91) and increasing work interval duration, while keeping work relief intervals constant, also increases T at $\dot{V}O_{2max}$ (136, 159, 181). For example, extending work interval duration from 30 to 60 s using a fixed-relief duration of 30 s in well-trained triathletes ($v\dot{V}O_{2max} = 19.9 \pm 0.9$ km/h) induced *very large* increases in T at $v\dot{V}O_{2max}$ (9 versus 1.5 min, $ES = +2.4$), despite a shorter total session time (28 versus 34 min, $ES = -0.9$; change in the T at $v\dot{V}O_{2max}$ /exercise ratio, $ES = +2.8$) (136). Similarly, in wrestlers ($v\dot{V}O_{2max} = 16.3 \pm 1.1$ km/h), Rozenek et al. (159) showed that increasing running work interval duration from 15 to 30 s led to *very large* increases in T at $\dot{V}O_{2max}$ (4 versus 0 min, $ES = 2.9$); a further increase in the interval duration to 60 s extended T at $\dot{V}O_{2max}$ to 5.5 min ($ES = 0.5$ versus the 30 s condition). Considering the importance of $\dot{V}O_2$ kinetics for extending T at $\dot{V}O_{2max}$ (137), these data suggest that longer work intervals (e.g., 30 s/30 s versus 15 s/15 s) are preferred for individuals with slow $\dot{V}O_2$ kinetics (i.e., older/less trained (143)) or for exercising on a bike (107).

Characteristics of the Recovery Interval Intensity and Its Impact on T at $\dot{V}O_{2max}$ During Short Intervals

In concert with the HIIT recovery bout discussion started in chapter 4, recovery bout intensity also plays a major role in the $\dot{V}O_2$ response during HIIT involving short intervals since it affects both the actual $\dot{V}O_2$ during the sets and exercise capacity (and hence, indirectly T_{lim} and T at $\dot{V}O_{2max}$; figure 5.11 (85, 174, 176)). Compared to passive recovery, runs to exhaustion involving active recovery are consistently reported to be 40% to 80% shorter (81, 82, 85, 174, 176). Therefore, when considering runs to exhaustion during 15 s/15 s exercises, the absolute T at $\dot{V}O_{2max}$ might not differ between active and passive recovery conditions (81) ($ES = -0.3$), but the T at $\dot{V}O_{2max}$ /exercise time ratio is substantially greater when active recovery is implemented ($ES = +0.9$), a factor of clear importance when implementing predetermined sets of HIIT (56, 136, 173). During a 30 s/30 s exercise model, compared with passive recovery, recovery intensities of 50% and 67% of $v\dot{V}O_{2max}$ were associated with *small* and *very large* increases in T at $\dot{V}O_{2max}$ ($ES = +0.4$ and $+0.1$ compared to passive recovery, respectively) and the T at $\dot{V}O_{2max}$ /exercise time ratio ($ES = +2.3$ and $+4.1$), respectively (174, 176). Increasing the recovery

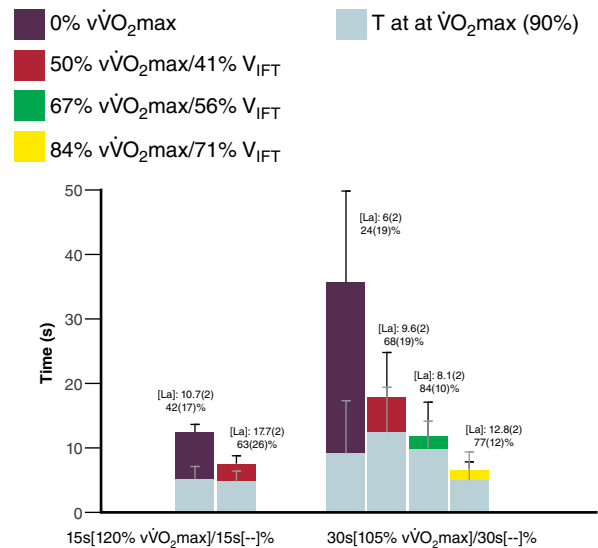


Figure 5.11 Mean $\pm$ SD total time and T at $\dot{V}O_{2max}$ during different short-interval HIIT as a function of changes in relief interval intensity. Percentages refer to the mean $\pm$ SD T at $\dot{V}O_{2max}$ relative to total session time. Mean $\pm$ SD [La] mmol/L is provided when available.

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intensity to 84% (the speed associated with the lactate threshold) *moderately* lowered T at $\dot{V}O_{2max}$ ($ES = -0.6$) but *very largely* increased the T at $\dot{V}O_{2max}$ /exercise time ratio ($ES = +3.4$).

In addition to limiting the drop in $\dot{V}O_2$ during the relief interval, which allows a greater $\dot{V}O_2$ to be reached during the subsequent exercise interval, active recovery also accelerates T to $\dot{V}O_{2max}$ (figure 5.12). T to $\dot{V}O_{2max}$ during different work and relief interval intensities involving short HIIT (30 s/30 s) was examined in young endurance-trained athletes (174-176). Not surprisingly, these studies showed shorter T to $\dot{V}O_{2max}$ values for work intensities $\geq 105\%$ $v\dot{V}O_{2max}$ and relief intensities $\geq 60\%$ $v\dot{V}O_{2max}$. Conversely, using slightly lower work and/or relief interval intensities (i.e., work 100% and relief 50% $v\dot{V}O_{2max}$) caused the runners to take more than 7 min to reach $\dot{V}O_{2max}$. Despite no statistical differences (174-176), all ES between the different work/relief ratios examined were *small* to *very large* (figure 5.12).

In the field then, T to $\dot{V}O_{2max}$ might be accelerated by manipulating HIIT variables during the first interval of the session, i.e., using more intense

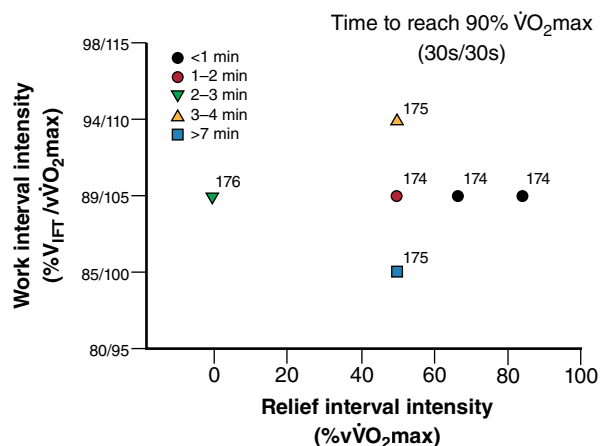


Figure 5.12 Time to reach $\dot{V}O_2\text{max}$ during HIIT with short intervals as a function of both work and relief interval intensities. Generally higher intensities of work and relief interval combinations speed the T to $\dot{V}O_2\text{max}$. (175); (176); (174). Numbers refer to references.

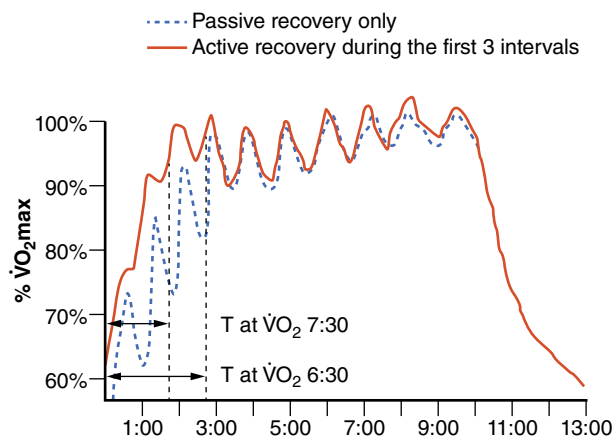


Figure 5.13 Effect of active recovery during the three first intervals only (the rest being passive) on $\dot{V}O_2$ during HIIT with short intervals.

work and/or relief interval intensities during the first 2 to 3 intervals or using longer work intervals and/or shorter relief intervals. This effect is illustrated in figure 5.13, where T to $\dot{V}O_2\text{max}$ was shown to be faster when active versus passive recovery was used during the first three intervals of a short-interval sequence. The fact that active recovery had a likely greater impact on T at $\dot{V}O_2\text{max}$ during the 30 s/30 s (174, 176) compared with the 15 s/15 s (81) exercise model is related to the fact that $\dot{V}O_2$ reaches lower values during 30 s of passive rest, which directly affects $\dot{V}O_2$ levels during the following effort.

In general, the characteristic of the recovery interval intensity can be aligned with the work intensity, with higher relief interval intensities used for lower work interval intensities (18) and lower relief exercise intensities used for higher work interval intensities and durations (83, 137, 175).

Series Duration, Sets, and T at $\dot{V}O_2\text{max}$ for Short Intervals

Dividing HIIT sessions into sets has been consistently shown to lower the total T at $\dot{V}O_2\text{max}$ (56, 136, 173). For example, in endurance-trained young runners ($v\dot{V}O_2\text{max} = 17.7 \pm 0.3$ km/h), performing 4-min recoveries (30 s rest, 3 min at 50% $v\dot{V}O_2\text{max}$, 30 s rest) every 6 repetitions (30 s/30 s) was associated with a *moderately* lower T at $\dot{V}O_2\text{max}$ ($ES = -0.8$) despite *very large* increases in T_{lim} ($ES = +4.3$). The T at $\dot{V}O_2\text{max}$ /exercise time ratio was therefore *very largely* reduced ($ES = -2.3$) (173). This is likely related to the time these athletes needed to return to high $\dot{V}O_2$ levels after each relief interval, irrespective of the active recovery used. While reviewing such studies could infer we advise athletes to consistently run short intervals to exhaustion to optimize T at $\dot{V}O_2\text{max}$, this would likely be challenging, psychologically speaking, for both coaches and athletes alike. This is likely why HIIT sessions to exhaustion are rarely practiced. In the field, the number of intervals programmed should be related to the goals of the session (total load or total T at $\dot{V}O_2\text{max}$ expected), as well as to T to $\dot{V}O_2\text{max}$ and the estimated T at $\dot{V}O_2\text{max}$ /exercise time ratio of the session.

If we consider that a goal T at $\dot{V}O_2\text{max}$ of ≈ 10 min per session is appropriate to elicit important cardio-pulmonary adaptations, athletes should expect to exercise for a total of 30 min using a 30 s [110% $v\dot{V}O_2\text{max}$]/30 s [50% $v\dot{V}O_2\text{max}$] format, since the T at $\dot{V}O_2\text{max}$ /total exercise time ratio is approximately 30%. While it's unrealistic to perform a single 30 min session, we can break such a session into 3 sets of 10 to 12 min (adding 1–2 min per set or series to compensate for the time needed to regain $\dot{V}O_2\text{max}$ during the second and third sets). This is a typical session used regularly by elite distance runners in the field. A lower volume (shorter series or less sets) is also used in other sports (i.e., in team sports, a T at $\dot{V}O_2\text{max}$ of 5–7 min is likely sufficient (35)) and/or for maintenance during unloading or recovery periods in an endurance athlete's program. In elite handball for example, $2 \times (20 \times 10$ s [110% V_{IFT}]/20 s [0] for a series duration of 10 min) is common practice, and might enable players to spend ≈ 7 min at

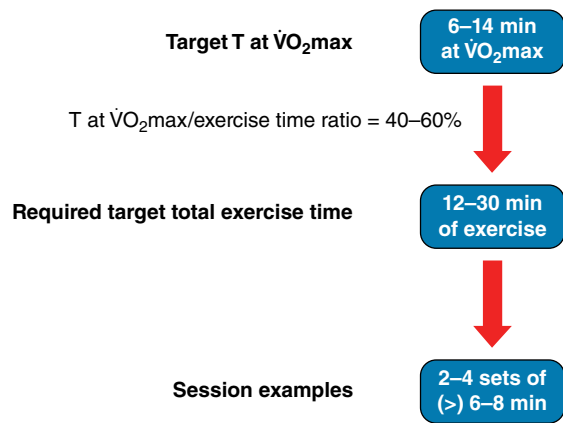


Figure 5.14 In the absence of best practice or scientific guidelines (15, 16, 21), the target total exercise time of short-interval HIIT can be defined using both T at $\dot{V}O_{2\max}$ and the T at $\dot{V}O_{2\max}$ /exercise time ratio. A good target T at $\dot{V}O_{2\max}$ will range from 6 to 14 min at $\dot{V}O_{2\max}$ with a T at $\dot{V}O_{2\max}$ /exercise time ratio of 40% to 60%. Total target exercise time will correspondingly range from 12 to 30 min.

$\dot{V}O_{2\max}$ (considering a T at $\dot{V}O_{2\max}$ /exercise time ratio of 35%, see chapter 27, Handball). In soccer, HIIT sessions such as $2 \times (12\text{--}15 \times 15 \text{ s } [120\% V_{\text{Inc.Test}}]/15 \text{ s } [0])$ are often implemented (80), which corresponds to ≈ 6 min at $\dot{V}O_{2\max}$ (14 min with a T at $\dot{V}O_{2\max}$ /exercise time ratio of $\approx 45\%$ (81)). See for example chapter 30, Soccer. It is however worth noting that throughout the years, Martin and some of his colleagues involved in elite team sports have had the tendency to decrease the volume of HIIT work with their athletes (e.g., $2 \times$ series of 4–5 min rather than $2\text{--}3 \times 6\text{--}10$ min as during their early careers), without any clear indication of a detrimental effect on physiological adaptations and performance. While only anecdotal, the minimum volume of HIIT required to elicit adaptations and improve performance in this population may be less than previously thought, and this should be a topic for future research.

A summary of the oxidative response recommendations for short intervals is shown in figure 5.14.

Short Versus Long Intervals and T at $\dot{V}O_{2\max}$

A direct comparison between long- and short-HIIT sessions, with respect to T at $\dot{V}O_{2\max}$, has to our knowledge only been reported twice in highly trained athletes. Gajer et al. (90) compared T at $\dot{V}O_{2\max}$ between 6×600 m (track session, 102%

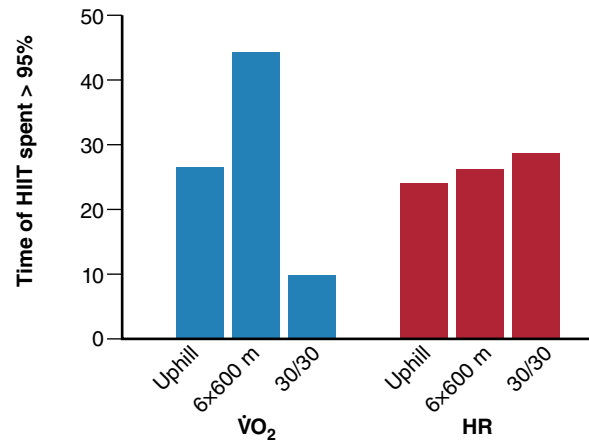


Figure 5.15 $\dot{V}O_2$ and HR responses during uphill and track HIIT with long intervals and HIIT with short intervals. Note: SD not provided by the authors. Note once more the dissociation between $\dot{V}O_2$ and HR.

Data from B. Gajer, C. Hanon, D. Lehenaff, and J.C. Vollmer, “Analyse comparée de différentes séances de développement de $\dot{V}O_{2\max}$,” in *Expertise et Sport de Haut Niveau: Actes des Entretiens de l’INSEP*, Novembre 2002, edited by D. Lehenaff and C. Mathieu (Paris: INSEP, 2003).

$v\dot{V}O_{2\max}$, run in ≈ 1 min, 40 s) and 10 repetitions of a 30 s/30 s HIIT session (work/relief intensity: 105/50% $v\dot{V}O_{2\max}$) in elite middle-distance runners ($v\dot{V}O_{2\max} = 21.2 \pm 0.6$ km/h). While $\dot{V}O_2$ reached 105% $v\dot{V}O_{2\max}$ during the track session, $\dot{V}O_{2\max}$ was not actually attained during the 30 s/30 s session (figure 5.15). If the track session is considered as the reference session (T at $\dot{V}O_{2\max}$ /exercise time ratio: 44%), T at $\dot{V}O_{2\max}$ was *very largely* lower during the 30 s/30 s intervals (10%, $ES \approx -2.6$). Similarly, Millet et al. (136) showed that performing 2 min/2 min intervals enabled triathletes to attain a *very largely* longer T at $\dot{V}O_{2\max}$ compared to a 30 s/30 s session ($ES = +2.2$, T at $\dot{V}O_{2\max}$ /exercise time ratio = +2.2). Long intervals were, however, *moderately* less efficient than the 60 s/30 s effort model (T at $\dot{V}O_{2\max}$ /exercise time ratio, $ES = -0.8$) (136). Taken together, these data suggest that long intervals and/or short intervals with a work/relief ratio > 1 should be preferred due to the greater T at $\dot{V}O_{2\max}$ /exercise time ratio, when focus on aerobic development is important.

Anaerobic Glycolytic Energy Contribution to Short Intervals

As with long intervals, in which the aerobic oxidative system is the primary physiological response apparent, there is still substantial involvement from

the anaerobic glycolytic energy contribution for short intervals. Thus, short intervals can be manipulated to target the anaerobic glycolytic energy component associated with type 3 and 4 acute responses, while lowered contributions enable the type 1 and 2 responses.

Work Interval Intensity and Duration

When we consider short-interval HIIT sessions (i.e., 15 s/15 s) of a similar mean intensity (i.e., 85% of $\dot{V}O_{2\max}$), perhaps not surprisingly, we find that higher work interval intensities elicit greater blood lactate responses and also shorten time to exhaustion (18). For example, blood lactate at exhaustion was 9.2 ± 1.3 mmol/L for work/relief intensities of 90/80% of $\dot{V}O_{2\max}$, 9.8 ± 1.4 mmol/L for work/relief intensities of 100/70% (ES=0.4 vs. 90/80%), and 11.3 ± 1.3 mmol/L for work/relief intensities of 110/60% (ES= 1.6 and 1.1 vs. 90/80% and 100/70%, respectively) (18). The effect of work duration on anaerobic contribution is also straightforward, with the longer high-intensity exercise periods eliciting larger draws on the glycolytic system. Interestingly however, as shown in figure 5.16, the repetition of very short efforts (i.e., 10 s) will in fact achieve a very limited engagement of anaerobic energy contribution, thanks predominantly to the use of the muscle's myoglobin stores. These myoglobin stores enable more reliance on the aerobic energy system during such very short duration intervals, and as a result, limit the anaerobic contribution.

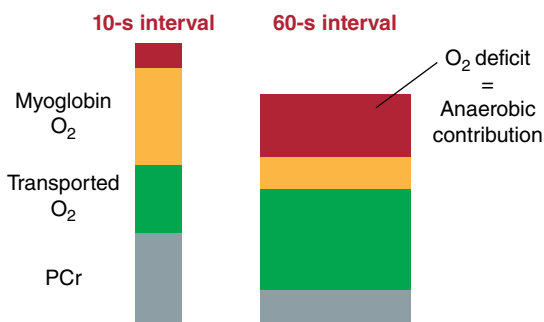


Figure 5.16 Theoretical illustration of the role that the stored oxygen in myoglobin plays during HIIT with short intervals of different durations, relative to the energy provided by transported oxygen (O₂) and phosphocreatine (PCr). The O₂ deficit (and resulting anaerobic energy contribution) is highly limited for very short intervals, during which the O₂ linked with myoglobin supplies most of the energy demands.

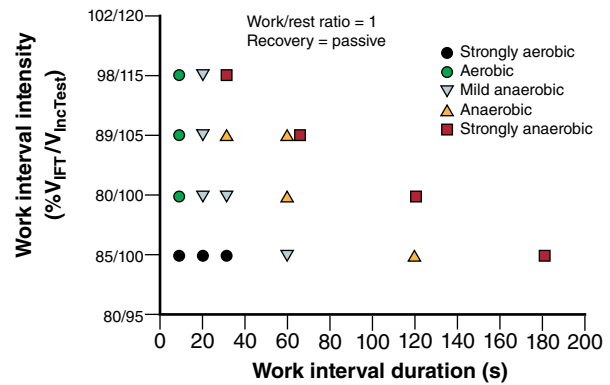


Figure 5.17 Schematic illustration of the energy system requirements for different forms of HIIT, with respect to blood lactate accumulation. Clearly, anaerobic involvement relates to higher work interval intensities and longer work interval durations. Strongly aerobic: 3 mmol·L⁻¹·5 min⁻¹; aerobic: 3 mmol·L⁻¹·5 min⁻¹; mildly anaerobic: 4 mmol·L⁻¹·5 min⁻¹; anaerobic: 5 mmol·L⁻¹·5 min⁻¹; strongly anaerobic: 6 mmol·L⁻¹·5 min⁻¹. T_{lim} time to exhaustion, $\dot{V}O_{2\max}$, minimal velocity associated with maximal oxygen uptake.

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The respective effects of work interval intensity and duration on blood lactate accumulation have also been examined in semiprofessional soccer players (peak incremental test running speed, $V_{Inc,Test}$ (figure 5.6) = 16.5 ± 2.3 km/h), during HIIT that included intervals lasting 10 to 30 s (17, 26, 76, 83, 133, 159, 162, 167, 180). Combined with data collected in other studies (e.g., see (55, 180)), this latter experimentation (26) shows how the selection of the appropriate combinations of different HIIT variables may be needed to reach specific blood lactate accumulation targets (figure 5.6). For exercise at 100% of $V_{Inc,Test}$, only work intervals longer than 1 min tend to be associated with high blood lactate levels. With work intensities of 110% $V_{Inc,Test}$, however, anaerobic glycolytic energy contribution is already likely increased when exercise is longer than 30 s. At a fixed work interval intensity, increasing the work/relief ratio is associated, not surprisingly, with substantial increases in the initial rate of blood lactate accumulation (figure 5.17) (159). Overall, however, with all data pooled from both long- and short-interval weapons (figure 5.6), it is clear that many short-interval formats will allow lactate to remain well below long-interval HIIT levels, which

may be the key advantage of using the short-HIIT weapon in certain cases (type 1 and 2 versus type 3 and 4).

Finally, it is worth noting that for extremely high exercise intensities (i.e., 20 s at 170% of $\dot{V}O_{2\max}$ interspersed with 10 s rest periods for 2 min), the accumulated O_2 deficit may reach maximum levels (172) (i.e., similar to the anaerobic capacity of the subjects, defined as the maximal accumulated O_2 deficit during an exhaustive 2-3 min of continuous exercise) (129) (chapter 3).

Glycogen Use With Short Intervals

As discussed in chapters 1 and 3, high-intensity exercise specifically draws on greater rates of carbohydrate oxidation, predominantly from a limited stored glycogen reserve. Its high use with HIIT is important to appreciate so as to appropriately manage (see factor 12 discussion in chapter 4). Figure 5.18 shows the glycogen degradation following different short-interval formats that all display the same intensity and work:rest ratio, differing only in their durations (7). Note from the figure how the longer exercise durations elicit higher blood lactate levels and larger decrements in muscle glycogen. Thus, longer short-interval formats will have a marked effect on both lactate production and the muscle glycogen use, and such training

needs to be managed and programmed appropriately (183, 184).

Recovery Period Intensity and Duration

As most authors examining the impact of the recovery period intensity during HIIT on blood lactate accumulation have used runs to exhaustion, which are not typically done in practice, the specific impact of the recovery interval intensity on blood lactate accumulation during actual field-based HIIT sessions performed by athletes is not clear (81, 174, 176). With increasing recovery intensity during a supramaximal 30 s (105% $\dot{V}O_{2\max}$)/30 s effort model, a progressive increase in blood lactate has been shown at exhaustion, despite a progressive decrease in exercise time (estimated from pH values: 6 ± 2 , 10 ± 2 , and 12.5 ± 2 mmol/L for recovery period intensities of 50%, 67%, and 84% of $\dot{V}O_{2\max}$, respectively) (174, 176). In contrast, compared to passive recovery, active recovery (40%-50% $\dot{V}O_{2\max}$) during a repeated submaximal 15 s (102% $\dot{V}O_{2\max}$)/15 s HIIT model has consistently been associated with *slightly* lower blood lactate values at exhaustion (10.7 ± 2.0 versus 11.7 ± 2.1 mmol/L, $ES = -0.5$ (81) and 12.6 ± 1.7 versus 13.1 ± 2.7 mmol/L, $ES = -0.2$ (85)). In the only study in which HIIT (15s [102% $\dot{V}O_{2\max}$]/15 s) was examined over comparable durations, active recovery

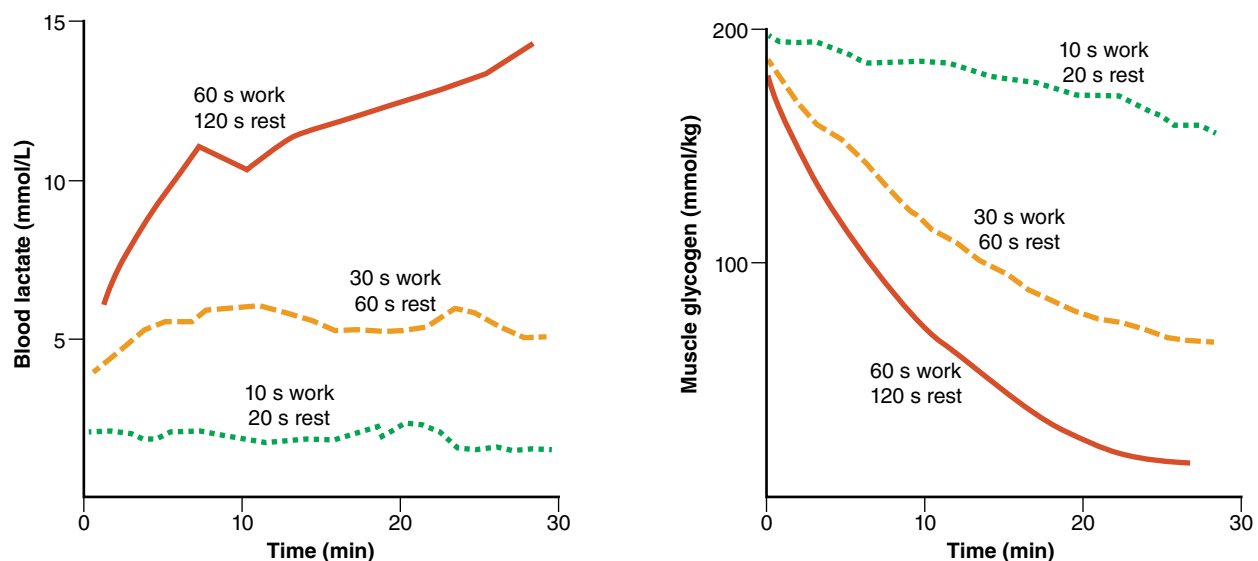


Figure 5.18 Lactate and glycogen levels during HIIT with constant work intensity, equivalent work:rest ratio, but varying bout durations.

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was shown to be associated with *moderately* higher post-HIIT blood lactate values (10.7 ± 2.0 versus 9.2 ± 1.4 mmol/L, $ES = +0.9$ (82)). If we consider that the lower muscle oxygenation level apparent with active recovery (85) (chapter 4) generally triggers anaerobic glycolytic system contribution during HIIT (69, 70), then an active recovery during short-interval HIIT is more likely to be associated with a *higher* anaerobic contribution (85). If we also consider that active recovery may be the preferred method for increasing T at $\dot{V}O_{2\max}$ during HIIT with short intervals, programming active-recovery HIIT sessions and achieving low blood lactate levels may be difficult. In practice, reducing the work/relief ratio and using passive recovery, as during a supramaximal HIIT format (e.g., 10 s [$>100\%V_{IFT}$]/20 s [0], holding V_{IFT} for the final speed reached at the end of the 30-15 intermittent fitness test, figure 5.6), provides an interesting alternative with respect to achieving both a high T at $\dot{V}O_{2\max}$ with moderate lactate production (i.e., type 3). In the latter case, the work interval duration prevents excessive anaerobic energy release yet is still great enough to achieve a high O_2 and the passive recovery duration allows for partial phosphocreatine (PCr) resynthesis (103), while limiting the drop in O_2 (figure 5.16). An alternative approach is to use a submaximal work interval intensity ($\leq 100\% v\dot{V}O_{2\max}$, which is less likely to trigger anaerobic glycolytic energy contribution) with active recovery periods ($\geq 50\% v\dot{V}O_{2\max}$) (24) (i.e., type 1). It is also worth noting that all field-based HIIT formats with short intervals are associated with low initial rates of blood lactate accumulation compared with long intervals (5, 68). As a result, the short-interval format may be a more sustainable, regular training weapon to use in the programs of some athletes, and less likely to elicit excessive sympathetic nervous system engagement that could lead to overtraining (chapter 7).

Work Interval Modality

The introduction of changes of direction (COD) into HIIT has been shown to moderately increase blood lactate accumulation, irrespective of the work intensity and duration ($ES \approx +1$, for all short-HIIT models tested, i.e., 10 s/10 s, 15 s/15 s, and 30 s/30 s) (74). This is not surprising given the increased mechanical demands of the repeated accelerations inherent with consecutive COD (148) and the fact that on average, the actual running speed is higher with COD (to compensate for the time lost while changing direction). In fact, COD running is likely to increase

peripheral (particularly biarticular locomotor muscles) and, in turn, systemic O_2 demands (with a possibly greater upper-body muscle participation). Therefore, while absolute anaerobic energy contribution is likely higher with COD (2, 104), the percentage contribution to total energy expenditure may be lower during high-intensity runs (47). When the intensity of the work intervals during HIIT was adjusted for the time lost with COD (figure 5.19), blood lactate values were surprisingly similar or even lower compared to HIIT without COD (37). The acute blood lactate responses to different short-interval HIIT formats with 10 s/20 s, including either running, sprinting, hopping, or squatting efforts every second work interval, were also compared in 8 highly trained adolescent handball players ($V_{IncTest}$ 16-17 km/h; figure 5.20) (72). Compared to the running-only condition, there were *very large* lower blood lactate concentrations following the sprinting ($ES = -3.5$) and squatting ($ES = -2.9$) conditions, but there was no clear difference with hopping ($ES = 0.0$). While the cardiorespiratory responses to these specific sessions should also be considered, these latter results provide relevant information for coaches wanting to manipulate the anaerobic glycolytic contribution to HIIT sessions within their periodized training plans.

Finally, running surface is another variable that likely affects anaerobic glycolytic energy release during HIIT. When team sport athletes (3 km running time = 12 min, 34 s) performed 3 sets of 7×25 to 45 s intervals performed at the highest sustainable intensity (W/R ratio 1/2 to 3) in the sand, their end-session blood lactate was largely greater ($ES = +1.0$) than when they did the same session on grass (25).

Neuromuscular Responses to Short-Interval HIIT Sessions

Short-interval HIIT can be manipulated to adjust for the degree of neuromuscular response. The neuromuscular response is emphasized in type 2 and 4 short intervals, while the response is attenuated in type 1 and 3 short intervals.

The acute neuromuscular responses to 5×300 m runs (77% of maximal sprinting speed [MSS], $\approx 120\%-130\% v\dot{V}O_{2\max}$) interspersed by 1 min recovery periods (100 m walk-jog) was examined in well-trained middle- and long-distance runners (164). Despite nonsignificant changes in maximal torque during a knee extensor MVC (-5% , with $ES \approx -1.2$, which still shows a large effect), the HIIT session caused severe acute peripheral fatigue, as

evidenced by reduced efficiency of excitation-contraction coupling (e.g., -28% for twitch torque, $ES > -5$). Muscle contractile function was recovered and even improved within 10 min following the session (e.g., $+11\%$ for twitch torque, $ES > 1$), while maximal torque during the MVC remained depressed for at least 120 min ($\approx -5\%$ – -6% , $ES < -1.2$). Due to the task dependency of acute muscular fatigue (86), the potential carryover effects of HIIT-related fatigue on subsequent training sessions involving sport-specific movement patterns (e.g., sprints, squats) is not straightforward and requires further research. In one study that assessed neuromuscular performance after an HIIT session (3 sets of 7×25 to 45 s intervals performed at the highest sustainable intensity, W/R ratio 1/2 to 3), most measures showed a return to baseline within 24 h (25). As well, the recovery of some measures was slightly better when the session was performed on sand versus grass. It is possible, therefore, that longer recovery periods may be required when sessions are performed on a harder surface (e.g., track, road). As well, consideration for the run surface and its influence on the neuromuscular load should be considered when programming run training.

While direct comparisons between long- and short-bout HIIT have yet to be documented, the acute neuromuscular load may be greater with short intervals for the following reasons.

1. Work intensity is generally higher with shorter intervals. While the majority of muscle fibers might already be recruited during long intervals (considering a minimal recruitment threshold at $>75\%$ – 85% $\dot{V}O_{2\max}$ for both type I and II fibers (4, 98)), the firing rate and relative force development per fiber is likely greater during short intervals (106).
2. Short intervals require frequent accelerations, decelerations, and reaccelerations (the occurrences of which are increased if the intervals are performed over shuttles). In addition to the increased metabolic and muscle force demands during acceleration phases of high-intensity exercise (78, 148), the completion of short intervals requires achievement of a greater absolute speed. For example, for a shuttle run completed at 120% of $\dot{V}O_{2\max}$ in the field, a small portion (i.e., 1–3 s) of such a run may in fact need to be run at 135% to compensate for the time lost during both the acceleration and deceleration phases (figure 5.21) (82). When this high-speed

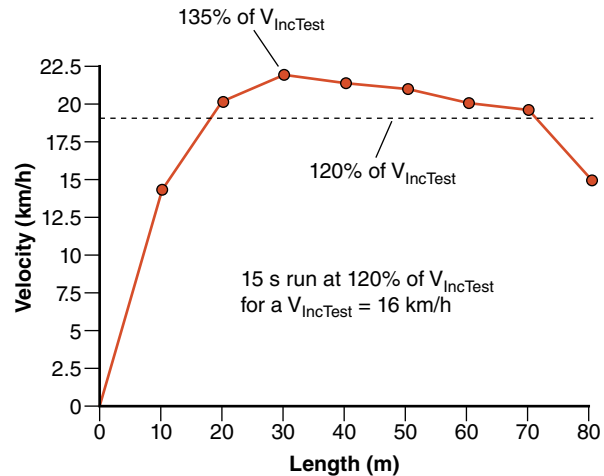


Figure 5.21 Example of the velocity versus distance relationship for a subject running 80 m in 15 s during a shuttle run (120% of $V_{IncTest}$, equal to 16 km/h). Portions of such a shuttle run actually need to be run at 135% of $V_{IncTest}$ to account for the required changes of direction, which has direct effects on neuromuscular load.

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portion of HIIT is considered in relation to the percentage of MSS and anaerobic speed reserve achieved ($>80\%$ and $>50\%$, respectively), the level of neuromuscular engagement is high (106) and should be considered within the context of training load and injury management (especially hamstring muscles (178) and traumatic-type injuries (88, 147, 166)).

3. For unique athletes, such as extremely tall or heavy basketball or rugby players, the musculoskeletal load of long intervals might actually be higher than during short intervals, due potentially to poor running technique and economy. Therefore, for these particular athletes, long run-based intervals should be avoided, or implemented on soft surfaces (e.g., sand, grass (25)), bikes, or rowing ergometers to prevent injuries.

While the specific impact of COD on the neuromuscular system during field-based HIIT sessions has yet to be examined in detail, it may exacerbate lower-limb muscular fatigue compared to straight-line running due to the additional accelerations and decelerations required (32). The impact of a strength-

oriented HIIT session on the neuromuscular system was also examined in 8 high-level endurance runners ($V_{\text{IncTest}} = 20.7 \pm 1.7$ km/h) (72, 77). Interestingly, when repeating six 200 m runs at 90%-95% of $\dot{V}\text{O}_2\text{max}$ (36-38 s) with alternate 30 s dynamic or explosive strength exercises, the runners avoided impairment in leg stiffness during running (8.08 ± 1.49 versus 7.87 ± 1.31 kN/m, $\text{ES} < -0.2$). While this might be related to the specific population of athletes (distance runners), this finding suggests that such training is well tolerated by distance runners and might have limited carryover effects on subsequent sessions. Whether injury risk is increased from such training is not known, but the increased load on the musculoskeletal system should be considered (77). Finally, the impact of different running patterns and exercise modes on acute neuromuscular responses during HIIT with short intervals was also examined in adolescent handball players (see chapter 27, Handball) (72). The acute neuromuscular responses to the four different HIIT sessions were protocol specific, with improvements in CMJ height shown following the running and squatting formats (possible postactivation potentiation (113)), and reductions in hopping height shown following the running and hopping conditions (possible neuromuscular fatigue related to localized overload on these muscle groups; figure 5.20). While the effect of these sessions on neuromuscular fatigue over longer durations (i.e., 24-48 h) is unknown, these data suggest that in team sport players, running patterns and exercise modes have important implications for programming, which should be considered with respect to other training sessions to avoid overload or injury (see chapter 7) and maximize adaptation.

Finally, in direct alignment with the anaerobic contribution, running surface has likely an acute effect on neuromuscular function. For example, it's reasonable to assume that softer surfaces (e.g., sand) will reduce impact force on the lower limbs compared to harder surfaces (e.g., grass or track (92)), which is likely to be of help for athletes returning from injuries or during periods of high training volume. Whether these acute loading differences substantially influence neuromuscular recovery in the following hours or days requires further research. Nevertheless, 24 h following intense short HIIT on the grass, these team sport athletes (25) moderately increased their CMJ and maintained repeated sprint performance, while after the same session performed in the sand, their CMJ was unaffected and their

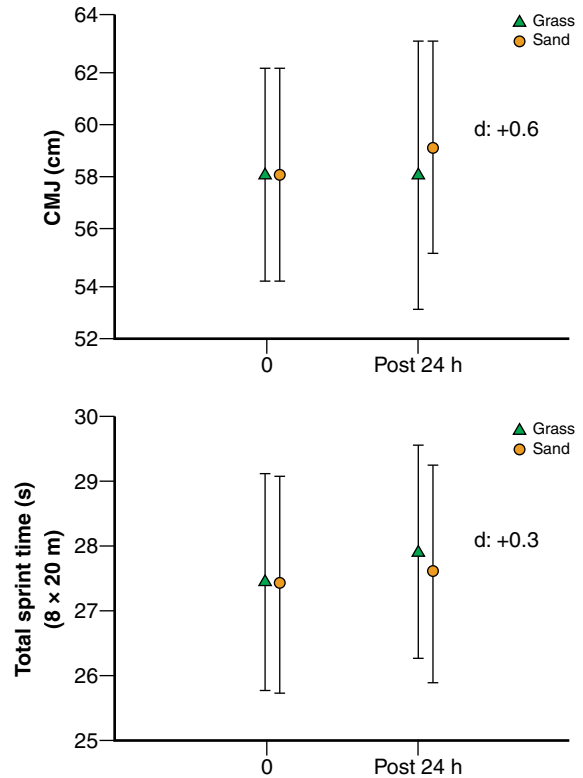


Figure 5.22 Change in countermovement jump (CMJ) and total repeated sprint time following HIIT with short intervals performed either on the sand or on grass (25).

repeated-sprint performance was slightly impaired (figure 5.22) (25).

Repeated-Sprint Training

Repeated-sprint training (figure 4.15), or a repeated-sprint sequence (RSS), is generally defined as the repetition of more than two short (≤ 10 s) all-out sprints interspersed with a short recovery period (< 60 s) (95). Recall that RSSs are used for type 4 and 5 targets, involving varying blends of acute oxidative, anaerobic glycolytic, and neuromuscular responses.

O₂ (Oxidative) Responses to Repeated-Sprint Sequences

To bring this section on RSS into proper context for the reader, it may be helpful to review figure 4.6, which shows the mean $\dot{V}\text{O}_2$ during six all-out 4 s running sprints in team sport athletes (44). Indeed, compared to the extensive data available on cardio-respiratory responses to long and short HIIT, relatively little has been presented on the acute responses

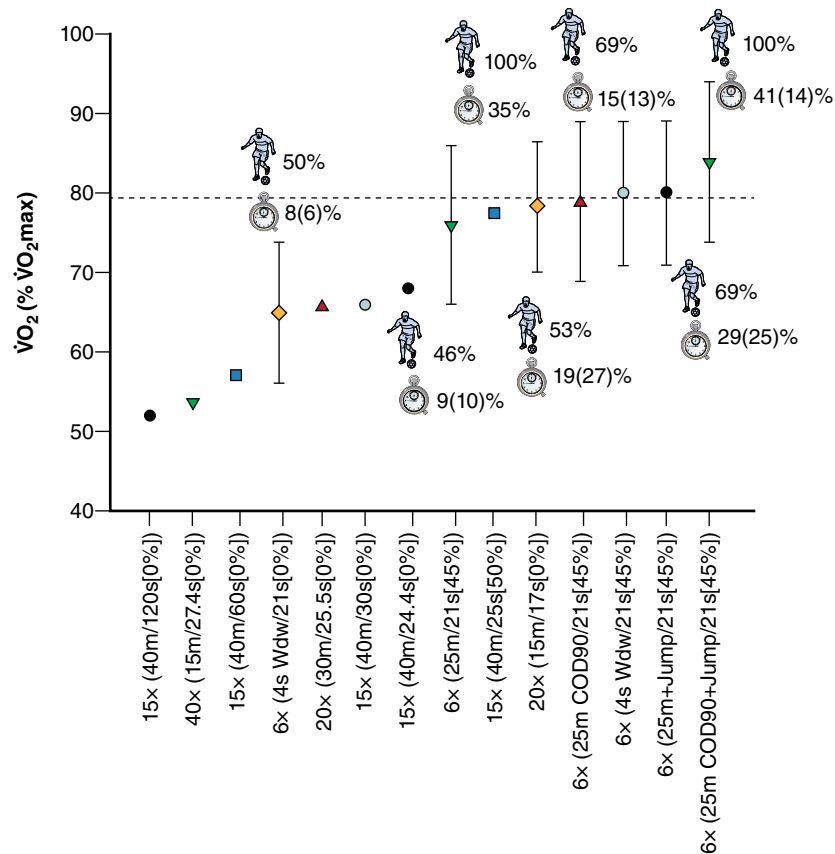


Figure 5.23 Mean $\pm$ SD $\dot{V}O_2$ responses during selected repeated-sprint sequences. The intensity of the relief interval (i.e., [percentage]) is expressed as a fraction of the $\dot{V}O_{2max}$. When available, both the percentage of participants that reached $\dot{V}O_{2max}$ (runner) and the mean $\pm$ SD time spent above 90% of $\dot{V}O_{2max}$ (clock) are provided.

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to RSS. Early in the 1990s, Balsom et al. (8, 9) demonstrated that RSSs were aerobically demanding and showed that this form of training can induce an aerobic response in excess of 65% $\dot{V}O_{2max}$. Dupont et al. (84) have also shown that soccer players can reach $\dot{V}O_{2max}$ with repeated sprinting. As shown in our two-part review, shown here in figure 5.23, we reanalyzed data from previous studies (36, 43, 44, 52) to provide T at $\dot{V}O_{2max}$ values for several forms of RSS (8, 9, 36, 43, 44, 52, 84). This analysis showed that when manipulating key variables already described (figure 2 in (33)), $\dot{V}O_{2max}$ is often reached and sustained for 10% to 40% of the entire RSS duration. If RSSs are repeated 2 or 3 times a session, as is often done (29, 57, 60), the majority of athletes will spend up to 2 to 3 min at $\dot{V}O_{2max}$ during the sprints.

As illustrated in figure 5.24, to increase T at $\dot{V}O_{2max}$ during an RSS, sprints or efforts should

last at least 4 s and recovery should be active and less than 20 s (8, 9, 36, 43, 44, 52, 84). The introduction of jumps following the sprints (36) and/or changes in direction (43) are also useful since these may increase systemic O_2 demand without the need for increasing sprint distance, which could lead to inappropriate muscular strain and/or greater injury risk. Nevertheless, with very short passive recovery periods (i.e., 17 s), some athletes can reach $\dot{V}O_{2max}$ by repeating only 3 s sprints (15 m). Nevertheless, with RSS, players do not always reach $\dot{V}O_{2max}$ and T at $\dot{V}O_{2max}$ shows high inter-individual variations (CV = 30%-100%) (84). For example, when considering the four different forms of RSS performed in the same group of 13 athletes (36, 43), it was observed that 6 athletes (45%) reached $\dot{V}O_{2max}$ on four occasions, 1 (8%) on three, 4 (31%) on two, and 2 (15%) never reached $\dot{V}O_{2max}$ during any of the RSSs. When

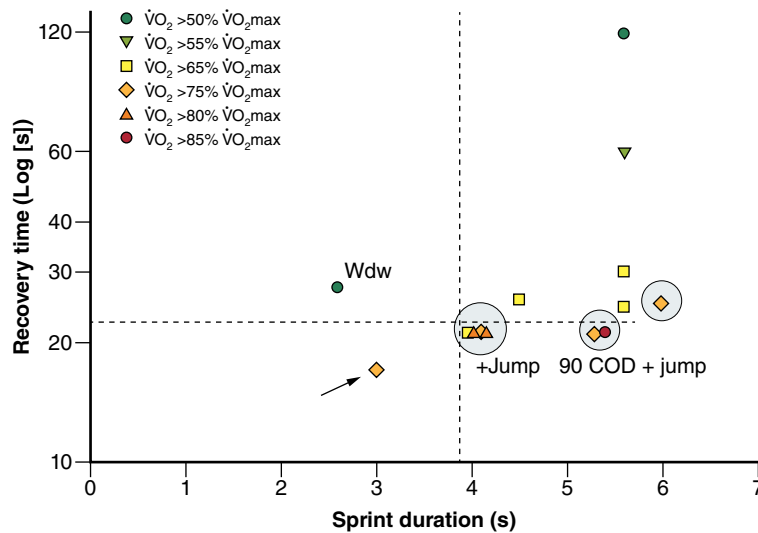


Figure 5.24 The mean $\dot{V}O_2$ responses during the repeated-sprint sequences presented are categorized into six color-coded families (based on the % of $\dot{V}O_2\text{max}$ elicited) and then plotted as a function of sprint and recovery duration. The dashed lines represent the shorter sprint and recovery durations likely needed to achieve at least 80% of $\dot{V}O_2\text{max}$. Circles indicate active recovery. Note (see arrow) that 75% of $\dot{V}O_2\text{max}$ can be achieved with very short sprints (i.e., 15 m) with passive recovery when sprints are interspersed with very short pauses (i.e., 17 s). COD: changes of direction; $\dot{V}O_2$: oxygen uptake; $\dot{V}O_2\text{max}$: maximal $\dot{V}O_2$; $v\dot{V}O_2\text{max}$: minimal running speed required to elicit $\dot{V}O_2\text{max}$. Wdw refers to sprints performed on a nonmotorized treadmill.

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the data from the four RSSs were pooled, the number of times that $\dot{V}O_2\text{max}$ was reached was inversely related to $\dot{V}O_2\text{max}$ ($r = -0.61$, 90% confidence limits, CL $(-0.84; -0.19)$). Similarly, the total T at $\dot{V}O_2\text{max}$ over the four different RSSs was inversely correlated with $\dot{V}O_2\text{max}$ ($r = -0.55$, 90%; CL $(-0.85; 0.00)$). These data indicate that, with respect to T at $\dot{V}O_2\text{max}$, the use in athletes of high fitness for targeting an acute oxidative response is questionable.

Anaerobic Glycolytic Energy Contribution During RSS

The all-out sprint nature of RSS naturally elicits a large anaerobic glycolytic contribution component in its acute response, and hits both type 4 and 5 targets. As shown in figure 5.25, end-exercise blood lactate values reported for RSS can range from 6 to 18 mmol/L (1, 8, 9, 36, 44, 46, 52, 64, 84, 125, 140) and can reach very high levels, i.e., similar to those reached with lactate production training (all-out 30 s efforts (115) or 300 m run at the speed maintained during a 400 m run ($v400\text{m}$) (99)). While methodological inconsistencies can partly explain the differences between studies, these data show that

manipulating the sprint distance or duration, the recovery intensity or duration, and work modality substantially impacts the anaerobic glycolytic contribution to an RSS.

Work:Relief Ratio

When sprints are longer than 4 s (i.e., >25 m) and when the recovery interval is less than 20 s and generally active, the initial rate of blood lactate accumulation is consistently high (i.e., $>10 \text{ mmol}\cdot\text{L}^{-1}\cdot 5\text{min}^{-1}$; figure 5.26). In contrast, shorter sprints and/or longer recovery durations can be less taxing on the anaerobic energy system. The initial rate of blood lactate accumulation during an RSS is also largely related to the work/relief interval, irrespective of the sprinting distance (figure 5.26). Interestingly, extremely high initial rates of blood lactate accumulation, that is, greater than those observed for all-out 30 s efforts (115) or 300 m run at 400 m sprinting speed (99), can be reached when repeating 4 s sprints on a nonmotorized treadmill (44) or when long sprints (>6 s) are separated by short recovery durations (≈ 17 s) (140). These observations are of particular interest for team sport coaches wanting to implement lactate production

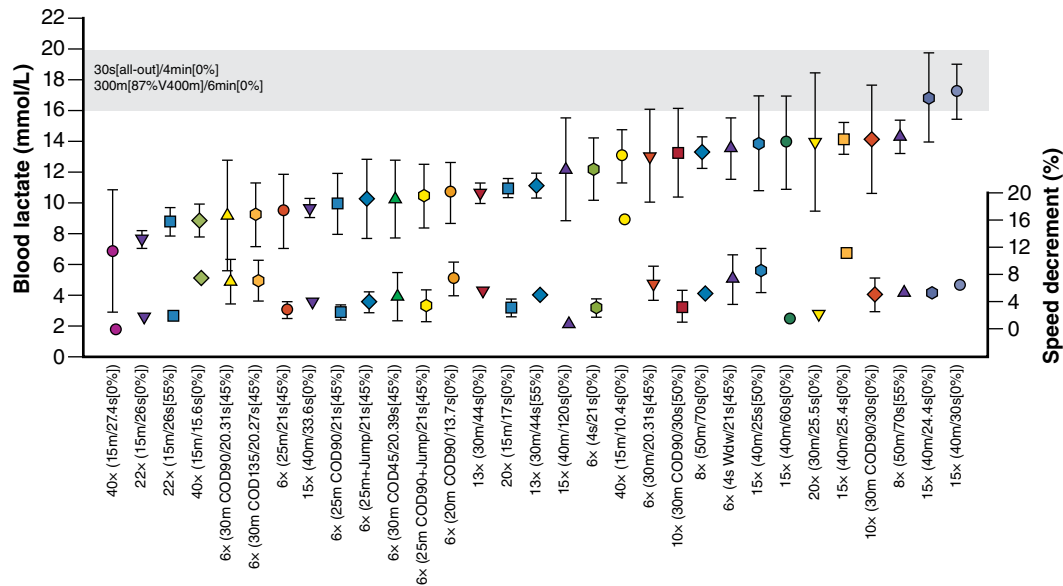


Figure 5.25 Estimated anaerobic energy contribution to different repeated-sprint sequence formats. Blood lactate measured following the different repeated-sprint sequences and their associated speed decrement (values are mean $\pm$ SD when available). The gray rectangular box represents the initial rate of blood lactate ([La]) increase during lactate production training, i.e., all-out sprints or at the speed maintained during a 400 m run (v400m).

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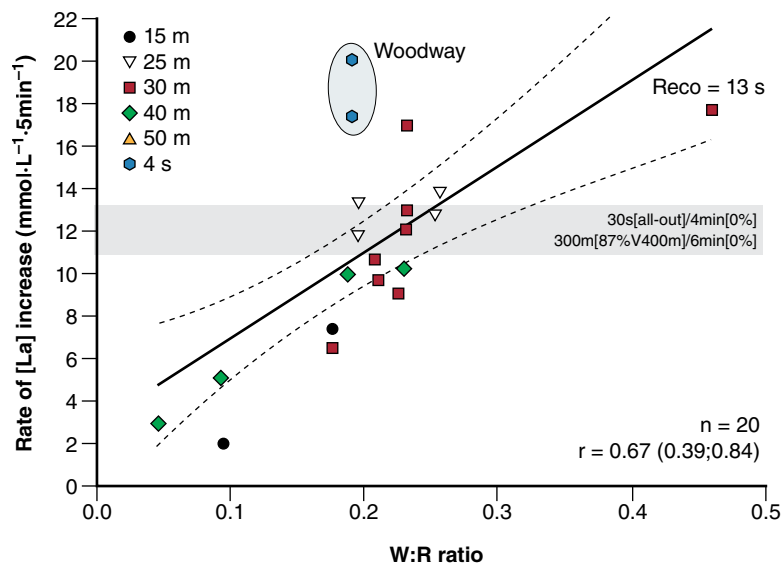


Figure 5.26 Relationships (r [90% confidence limits]) between the initial rate of blood lactate ([La]) increase during select repeated-sprint sequences and work:rest (W:R) ratios. Correlation coefficients are provided with 90% confidence intervals. The gray rectangular box represents the initial rate of blood lactate increase during lactate production training, i.e., all-out sprints or at the speed maintained during a 400 m run (v400m). The black oval highlights the two sprints performed on a nonmotorized treadmill (i.e., Woodway).

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training since RSSs are more team sport specific and might improve motivation in players reluctant to train on a track and/or perform longer sprints.

Work Modality

In addition to the work:recovery ratio, incorporating COD and/or jumps into RSS has the potential to influence the blood lactate response. When 25 m sprints (departing every 25 s, relief intensity = 7 km/h) are repeated over a shuttle run (180° COD), blood lactate can be increased (9.3 ± 2.4 versus 10.0 ± 10.7 mmol/L, $ES = +0.3$) (43). However, when RSSs are matched for initial sprint time (requiring a reduction in sprinting distance as COD angles increase), RSS with or without 90° COD angles are recommended to either minimize or maximize blood lactate accumulation, respectively (46). For RSSs that involve 0°, 45°, 90°, or 135° COD angles, $\Delta[\text{La}]$ (post-resting values) of 10.1 ± 2.2 , 8.0 ± 2.3 , 6.1 ± 2.5 , and 7.4 ± 2.3 mmol/L have been reported, with *moderate-to-large* differences between those COD angles shown (except 45° and 135°). While it's intuitive to think that muscle recruitment (and blood lactate) increase with larger COD angles, associated absolute sprinting speed would be lower in this particular setting (due to the shorter distances) (46) (figure 5.27). Finally, jumping after each sprint is also associated with *small* increases in blood lactate accumulation ($ES = +0.2-0.3$) (36),

probably as a consequence of the increased total muscular work (figure 5.28).

Neuromuscular Responses to RSS

As with the anaerobic glycolytic component, the all-out nature of RSS naturally elicits large demands on the acute neuromuscular response, enabling us the ability to reach both type 4 and 5 targets. During RSS, the reduction in running speed observed reflects the progressive increase in overall locomotor stress, as evidenced by an impaired force production, changes in stride patterns (lower stride frequency and stride length), decreased musculoskeletal stiffness (96), and both neuromuscular adjustments and metabolic disturbances at the muscle level (95). Despite its poor reliability (145), the percentage of speed decrement (%Dec) is still the more commonly reported index used to assess acute fatigue during RSS, a marker that varies from 1% to 12%, depending on the RSS format (figure 5.29).

With respect to neuromuscular load, %Dec values should be interpreted with caution. While a high %Dec is likely associated with an increased (muscle) fatigue in the acute setting, the actual musculoskeletal strain of the sequence (with respect to possible muscle damage and/or injury risk) is more likely related to running patterns and the running speed maintained during each repetition (i.e., individuals with higher running speeds and low %Dec may have

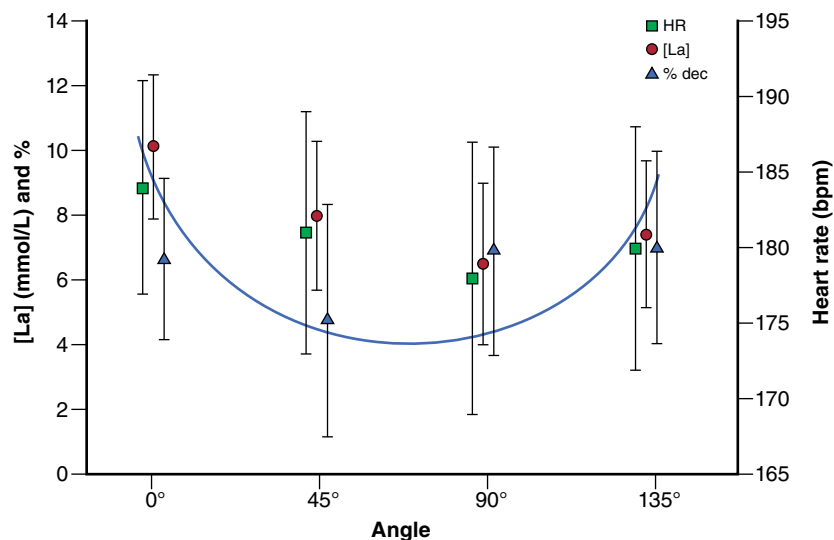


Figure 5.27 Heart rate (HR), blood lactate ([La]), and speed decrement (%Dec) responses to repeated-sprint sequences including straight-line (0°), 45°, 90°, or 135° change-of-direction sprints.

Data from M. Buchheit, B. Haydar, and S. Ahmaidi, "Repeated Sprints With Directional Changes: Do Angles Matter?" *Journal of Sports Sciences* 30 no. 6 (2012): 555-562.

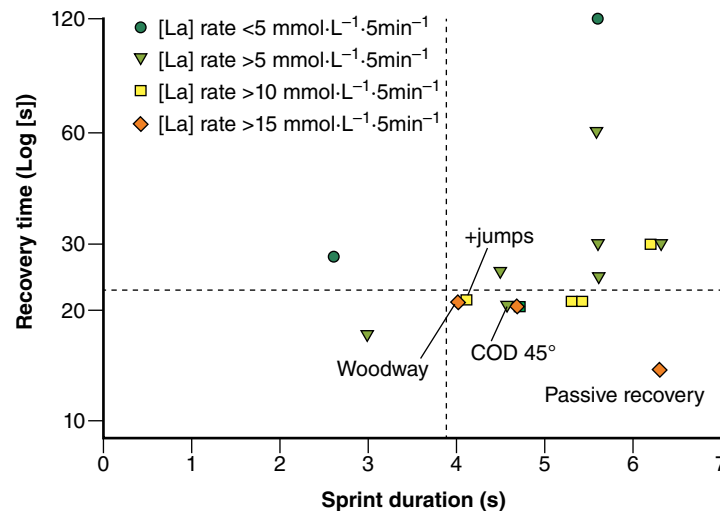


Figure 5.28 Rate of blood lactate increase during different repeated-sprint sequences as a function of recovery and sprint duration. The dashed lines represent the shortest sprint and recovery duration likely needed to achieve a rate of blood lactate increase (10 mmol·L⁻¹·5 min⁻¹) with the exception of one repeated-sprint sequence performed with 45° changes of direction (COD) (5 mmol·L⁻¹·5 min⁻¹). All repeated-sprint sequences leading to a rate of blood lactate increase (10 mmol·L⁻¹·5 min⁻¹) included an active recovery, except for one with passive recovery. COD: changes of direction, with associated angle; Woodway refers to sprints performed on a nonmotorized treadmill.

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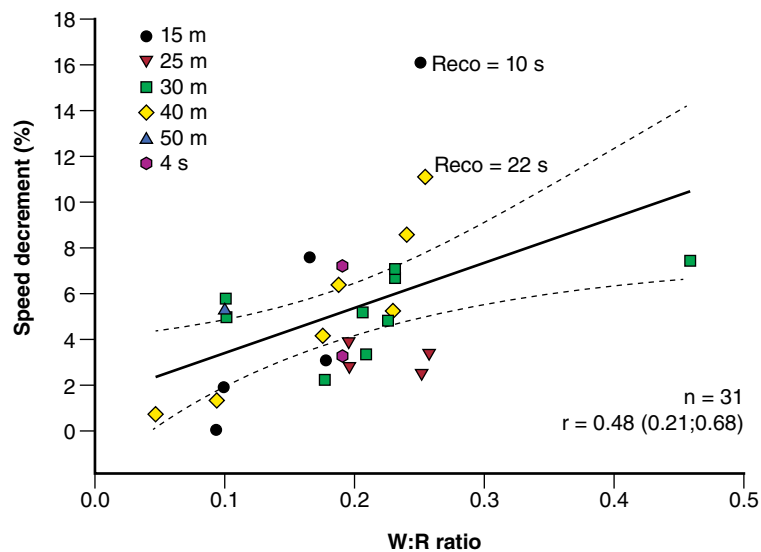


Figure 5.29 Relationship (r [90 % confidence limits]) between the speed decrement of various repeated-sprint sequences and work:relief (W:R) ratios. Correlation coefficients are provided with 90% confidence intervals.

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a high musculoskeletal strain). When data from these studies were pooled, %Dec was moderately and positively correlated with the work:relief ratio ($r=0.48$, 90% confidence limits (0.21; 0.68), figure 5.29). A higher work:relief ratio is generally associated with reduced PCr resynthesis and an accumulation of blood lactate (figure 5.26) and metabolites in the muscle, which may partially explain the greater impairment of repeated-sprinting capacity (95). It is, however, worth noting that for extremely short recovery periods (i.e., 10 s (125)), the percentage of speed decrement is dramatically increased (see outlier data point, figure 5.29). Since this latter format might impose a substantial load on the musculoskeletal system, it should be implemented only with consideration for the timing of the other neuromuscular-oriented training sessions to avoid overload.

In addition to variations in sprint duration and work:relief ratio, the introduction of COD to RSS can also affect the fatigue profile and hence acute neuromuscular load. Compared to straight-line sprinting, for example, RSS with 180° COD were associated with a *slightly* lower %Dec ($ES=-0.4$) (43). When comparing the effect of COD angles *per se* (on sprints adjusted for initial sprint duration), the fatigue response to the RSS has been shown to be angle dependent (46). Values for %Dec were 6.7 ± 2.5 , 4.8 ± 3.6 , 7.0 ± 3.2 , and $7.1 \pm 3.0\%$ for straight line, 45°, 90°, and 135°, respectively (with %Dec for 45° being substantially lower than for the three other conditions) (46) (figure 5.29). Running with 45° COD required variation in muscle activation compared to straight-line running (slowing down slightly to turn), without the need to apply large lateral forces, such as with the greater angles (135°) (32). This finding suggests that repeated sprints involving 45° COD may be an effective alternative to reducing acute neuromuscular load during RSS. Another important point to consider when it comes to introducing COD into RSS is the likely modification in lower-limb neuromuscular control. In fact, compared to repeated straight-line high-intensity 16.5 m runs, runs with 90° COD (performed within 4 s, departing every 20 s) were associated with both an increased stabilization time after a single-leg drop jump and a selective reduction of electromyography activity in hamstring muscles (101). Therefore, due to potential transient loss of ankle and knee stability, RSS including sharp COD might expose athletes to higher risk of ankle sprain and knee injuries, especially for athletes not used to performing such movement patterns at high speed (e.g., combat sport athletes,

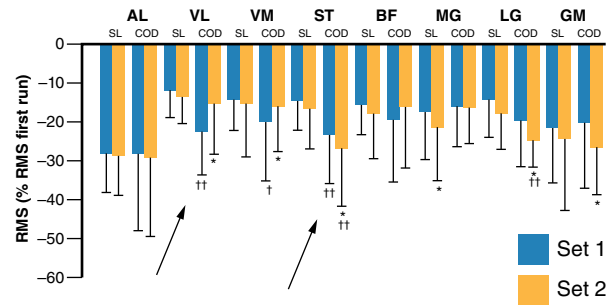


Figure 5.30 Changes in electromyography amplitude (RMS) for eight muscles during and after high-intensity intermittent efforts, with (COD) or without (i.e., straight line, SL) changes of direction. *: possible within-condition difference versus first set; †: possible difference versus SL; ††: likely different. AL: adductor longus; VL: vastus lateralis; VM: vastus medialis; ST: musculus semitendinosus; BF: biceps femoris; MG: medial gastrocnemius; LG: lateral gastrocnemius; GM: gluteus medius. SL: straight line; COD: change of direction; $n=11$.

gymnasts). In practice, in addition to modulating the metabolic aspects reviewed above, changing direction during RSS might be an effective training practice to promote long-term COD-specific neuromuscular adaptations aimed at improving performance and knee joint stability (figure 5.30).

The specific impact of deceleration during RSS has the potential to increase acute muscle fatigue (i.e., impaired repeated sprinting performance), but only when a large number of sprints are performed, i.e., >11 sprints (119). Finally, adding jumps after each sprint during RSS is also likely to increase neuromuscular load. This was shown by a *moderately* greater %Dec for sprints, with ($ES=+0.7$) and without ($ES=+0.8$) shuttles (180°) (36). Finally, when jumps were added to RSS that involved shuttles, the %Dec for jump performance was highest (12 ± 4 versus $8 \pm 4\%$ for shuttle versus straight sprints, $ES=+0.8$), which suggests that this latter RSS format is likely more demanding on lower limbs. With respect to traumatic injuries, the introduction of COD during RSS has the advantage of restricting sprinting distance and stride length, which might help prevent hamstring overload and acute injuries. In contrast, as for HIIT with short intervals, RSS including COD might expose athletes to a higher risk of ankle sprain.

Finally, the time needed to recover from acute neuromuscular fatigue following RSS is important

for RSS programming (158). In cycling studies, it has been shown that when a second RSS was performed 6 min after the first, fatigue was accelerated (i.e., %Dec of 17 versus 13% for the second versus the first set) (131), suggesting that a certain level of neuromuscular fatigue remained. When interspersing run-based RSS with 5 min rest, followed by 6 min active recovery and then a short specific warm-up (total time $\approx$ 15 min), repeated-sprint performance was not impaired during the second set (36). Explosive strength, however, had not fully recovered, since CMJ height was 5% lower ($ES = -0.4$) than before the first set (36). Compared to cycling, however, changes in neuromuscular coordination and stride adjustments appear to compensate for the acute fatigue during sprints to maintain running performance (160). This data suggests that when RSSs are repeated within a short time period (i.e., within 2 to 5 min (29, 57, 60)), neuromuscular fatigue, as evaluated from post-exercise jump tests, is heightened. To implement quality RSS sessions (if this is indeed possible (38)), a prolonged and likely active recovery period should be implemented between sets (possibly >15-20 min) to maximize muscle recovery.

Sprint Interval Training

Sprint interval training (SIT, figure 4.16), also termed speed endurance training, is another form of HIIT that involves near to maximal (all-out) efforts (115). Compared to the RSS format, efforts and recovery periods are typically longer (e.g., repeated 30 s sprints interspersed with 2-4 min of passive recovery (115)). SIT is similar to RST or RSS in that both are performed all out, so they can be prescribed without the need to pretest the individual (i.e., $v/p\dot{V}O_{2max}$ is not specifically needed to calibrate the intensity). SIT differs from RST/RSS only in terms of the time range. RST/RSS could be described as short sprints, and tend to range from 3 to 10 s, while SIT is sometimes called long sprints, and usually last from 20 to 30 s in duration (94, 115). SIT is exclusive to type 5 targets in terms of their acute responses, requiring predominantly anaerobic glycolytic and neuromuscular responses.

$\dot{V}O_2$ Responses to Sprint Interval Sessions

As with our understanding of the RSS in the previous section, it may be prudent to review the all-out sprint training background in chapter 2, and specifically

with reference to figure 2.10, which shows the power output, $\dot{V}O_2$, and muscle tissue oxygenation index response during a SIT session. Notwithstanding the important research that demonstrates its benefits (94), only a small number of studies show the acute physiological responses to typical SIT sessions that might be implemented in practice. Tabata et al. (172) showed that $\dot{V}O_{2max}$ was not reached (peak of 87% O_{2max}) during repeated 30 s cycling efforts (at 200% $p\dot{V}O_{2max}$, and therefore not actually all-out) interspersed by 2 min passive recovery. In contrast, we have shown (41) during a fully all-out SIT session that most athletes reached values close to (or above) 90% of their $\dot{V}O_{2max}$ and HR_{max} (figure 2.10). In this study, T at $\dot{V}O_{2max}$ was only 22 s on average (range 0 to 60 s, with two athletes not reaching >90% $\dot{V}O_{2max}$) (41), and $\dot{V}O_{2max}$ was reached by only five subjects (50%). These important individual $\dot{V}O_2$ responses to SIT sessions were partly explained by variations in cardiorespiratory fitness (i.e., there was a negative correlation between T at $\dot{V}O_{2max}$ and both $\dot{V}O_{2max}$ ($r = -0.68$; 90% CL $(-0.90; -0.20)$) and $\dot{V}O_2$ kinetics at (submaximal) exercise cessation ($r = 0.68$; 90% CL $(0.20; 0.90)$). As with RSS, there was no link between T at $\dot{V}O_{2max}$ and $\dot{V}O_2$ kinetics at exercise onset. Finally, it is worth noting that although pulmonary $\dot{V}O_2$ is not high during SIT, muscle O_2 demand is, especially as the number of sprint repetitions increase (figure 2.10). It has been shown that there is a progressive shift in energy metabolism during a SIT session, with a greater reliance on oxidative metabolism when sprints are repeated (27, 150). Along these same lines, muscle deoxygenation levels and post-sprint reoxygenation rates have been shown to become lower and slower, respectively, with increasing sprint repetition number. This response implies a greater O_2 demand in the muscle with increasing sprint repetition, since O_2 delivery is likely improved with exercise-induced hyperemia (blood excess in muscle bed supplying vessels) (41).

Anaerobic Glycolytic Energy Contribution During SIT

As with RSS, the all-out nature of SIT naturally engages a high anaerobic-glycolytic response. When physical education students repeated 4×30 s sprints at 200% of $p\dot{V}O_{2max}$ interspersed with 2 min rest periods, the accumulated O_2 deficit reached 67% of their anaerobic capacity (172), defined as the maximal accumulated O_2 deficit during an exhaustive

2 to 3 min continuous exercise (129). Total accumulated O_2 deficit over the four sprints, however, was three times higher than the anaerobic capacity. Blood lactate levels generally reach 16 to 22 mmol/L (28, 42, 115, 128), which corresponds to blood lactate accumulation rates >10 to $15 \text{ mmol} \cdot \text{L}^{-1} \cdot 5 \text{ min}^{-1}$ (figure 5.26). While few researchers have manipulated SIT variables to examine their effect on the anaerobic glycolytic energy contribution, it is expected that slightly shorter sprints and/or lower intensities would reduce its input. As well, sprints >45 s are more likely associated with a greater contribution from the aerobic system (i.e., $>40\%$ (91)), which lowers anaerobic glycolytic energy demand in ensuing sprints. Thus, a general rule is that recovery periods should be long enough (i.e., >1.5 -2 min) to allow for the aerobic system to return to resting levels, and thus enable an O_2 deficit to occur at the onset of the following exercise (160). Since an increased intramuscular H^+ concentration may inhibit glycolysis (89, 155), extending the recovery period might also allow for subsequent sprints to be achieved with a higher acid/base status, allowing, in turn, a greater anaerobic glycolytic energy contribution. For instance, post-exercise blood lactate levels were shown to *moderately* increase from 13.3 ± 2.2 to $15.1 \pm 1.7 \text{ mmol/L}$ ($ES = +0.9$) in well-trained team sport athletes ($V_{IFT} = 18.9 \pm 1.5 \text{ km/h}$) when the recovery between 30 s all-out shuttle sprints was increased from 30 to 80 s (45). Maximal blood lactate values were also reported to be greater when 30 s sprints were interspersed with 4 min of recovery (≈ 17 (28) and 22 (127) mmol/L) compared to 2 min ($15.3 \pm 0.7 \text{ mmol/L}$) (41). Finally, in elite Spanish 400 m runners (92% of the world record times) performing 300 m sprint repetitions (run in 40-45 s with varying inter-sprint recoveries), blood lactate after the sixth sprint was higher for sessions run at 87% of v_{400m} ($\approx 75\%$ -80% of MSS) with 6 min recovery (17-19 mmol/L) compared to sessions involving sprints at 91% of v_{400m} ($>80\%$ MSS) but separated by only 3 min of recovery ($\approx 10 \text{ mmol/L}$) (99). Again, these results show how SIT variable manipulation can lead to different blood lactate levels, which has direct implications for HIIT programming.

Neuromuscular Responses to SIT

As with the anaerobic glycolytic system, the all-out nature of SIT additionally will engage a large response from the neuromuscular system. The large speed or power decrement score generally observed (20% for

repeated 30 s cycling sprints (41, 127), 5%-20% for track sessions involving repeated 300 m runs (99)) suggests that neuromuscular function is largely impaired following a SIT session. Studies suggest that in contrast to other forms of HIIT (e.g., short duration and/or nonmaximal efforts (120, 151)), central mechanisms may be the primary origin of the impairment to MVC performance following long sprint repetition (87). There is, however, little data available examining the neuromuscular response to SIT variable manipulation. Neuromuscular fatigue (as assessed by jump height) following an HIIT session in elite 400 m runners (personal best = 92% of world record) is likely protocol dependent (99). The magnitude of the SIT-induced reduction in CMJ height was shown to be positively correlated with initial CMJ height and is affected by between-run recovery duration, with longer recovery durations (and hence the greater the anaerobic glycolytic energy contribution) showing greater impairments in jump height (99). When examining the neuromuscular responses in team sport athletes to single maximal sprints lasting 15 s (100 m), 31 s (200 m), and 72 s (400 m), Tomazin et al. (177) showed that knee extensor MVC reduction was apparent immediately and 5 min after the 72 s sprint and was fully recovered 30 min post-sprint.

In run-based sports, SIT is often implemented in a straight-line fashion or over an arcing shape of >200 m (running over half of a soccer pitch, for example). While this setting can allow players to reach very high running speeds (close to their MSS for the first intervals), it can dramatically increase hamstring injury risk (178). SIT sessions should therefore be considered using a cost-benefit approach. It is clear that SIT is unlikely to develop MSS (38, 58). However, when an appropriate warm-up is used and when distance is built up over the sessions, such SIT formats can, in addition to triggering the well-known metabolic adaptations in muscle (94), serve as a prophylactic intervention (i.e., preparing lower-limb muscle and tendons to tolerate the extreme tensions associated with maximal sprinting). For a safer option (or for the initial sessions of a SIT program), coaches can implement SIT sessions using 40 m shuttles (58).

Game-Based High-Intensity Interval Training

Recall from chapter 4 that game-based HIIT (GBHIIT) involves an aspect of play and decision-making when

the format is being conducted. While much research has been published in the area of GBHIIT, and specifically team sport-based SSGs (with the great majority published in soccer) (112), it is beyond the scope of our book to exhaustively summarize all findings, since the effects of SSG-related variable manipulation are sport-, population-, and context-specific. We have therefore opted to present the main trends observed (but not the magnitude of the effects, as we have done for run-based HIIT) so as to illustrate examples of how this specific puzzle piece can be shaped, mainly with data arising from the soccer context. It is also worth noting that because of the inherent tactical and technical components of SSGs, many team sport practitioners program these using more tactical rather than physiological considerations (e.g., chapter 23, Basketball). There is obviously no right or wrong way to do this. For example, Martin still programs GBHIIT using physiology-based objectives (see soccer and handball chapters) and for consistency with the other chapters, we have kept the structure of this weapon similar to the others so as to be consistent in describing its acute physiological responses. Indeed, GBHIIT can be used to hit type 2 through 4 acute target responses.

GBHIIT is an HIIT format that differs substantially from the other four weapons for several reasons:

1. With the other weapons already presented (figures 4.13 to 4.16), exercise mode is typically an option. With GBHIIT, however, work modal-

ity must be sport specific by default, i.e., athletes practice their own sport under specific rules and contexts (e.g., number of players, specific areas).

2. Work intensity can't be prescribed as a function of given physiological entities such as $V/P_{IncTest}$, V_{IFT} , or CP/CV . Therefore, in contrast to the four other formats, exercise intensity is more affected by the manipulation of GBHIIT-specific variables (e.g., player number, field size, rules), which indirectly affects a player's effort and subsequent intensity (figure 5.31).
3. The important skill or tactical component is a two-edged sword. While it creates a fantastic opportunity to reinforce and even develop key aspects of team sport, together with a metabolic stimulus (table 5.1, figure 5.32), the highly variable quality and density of player interactions lead to a more variable and less predictable metabolic response, which can be limiting when a specific biological adaptation is being targeted.

Table 5.1 provides a thorough description of the advantages and disadvantages of using SSGs over traditional run-based HIIT in team sports.

Stratton et al. (171) have developed a simple model to characterize sport activities, with the view of enhancing skill acquisition and motor learning. The model is based on the respective levels of variability and randomness of game situations, and assumes that the variability within a practice environment

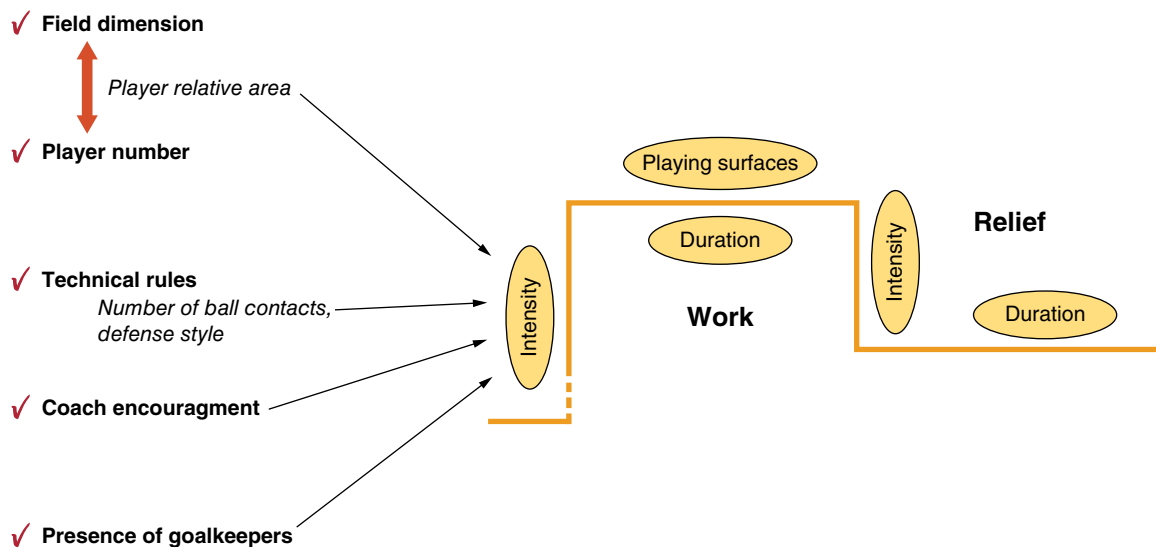


Figure 5.31 Methods used to modulate exercise intensity during GBHIIT.

Table 5.1 Advantages and Disadvantages Associated With Physical Training Using Soccer Drills and Generic Running

Soccer drills	Traditional running
A small-sided game with specific constraints (scoring, targets, balls, players, rules, and area)	A controlled running session with players exercising for a specific time or distance
Advantages	
Improved motivation	Exact work intensity can be easily controlled
Enhanced training of movement efficiency	Improvements can be monitored objectively
Improvements in tactical awareness	Comparisons can be made between players
Improvements in technical skill	Gain insight into player character or motivation
Optimizes training time and physical load	
Potential decrease in injuries	
Disadvantages	
Exact work intensity is difficult to control	Less movement associated with match play
Often difficult to organize optimal training structure	Players do not practice technical skills
Increased risk of contact injuries and overuse injuries (repeating over and over the same movement patterns)	No game-based tactical elements
Need numbers to make up session	Players do not like running
Certain degree of technical ability required	May increase risk of some injuries (tendonitis, lumbopelvic problems) due to unaccustomed running
Possible ceiling effect for very fit players	

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elicits a more flexible generalized motor program, allowing the natural variety of similar (but different) situations encountered in competition. The effects of such variable practice may also enhance an athlete's performance potential outcome due to the random versus blocked (predictable) practice conditions enabled. In a blocked practice condition, players are required to practice a single skill such as passing, shooting, or hitting during the session. In contrast, under random (game-based) practice conditions, skills are performed randomly, in an ad hoc manner, as they must be in the competition environment. While blocked practice may be better for short-term performance, random practice conditions might enhance skill-based learning. Figure 5.32 illustrates where traditional HIIT and SGG formats reside within this classification, and illustrates the importance of GBHIIT for team sport and racket-based athletes, irrespective of their metabolic responses.

VO₂ Responses to GBHIIT

Despite its important use in the field, only a few investigations have actually measured $\dot{V}O_2$ during GBHIIT. This is mainly due to both logistics (e.g.,

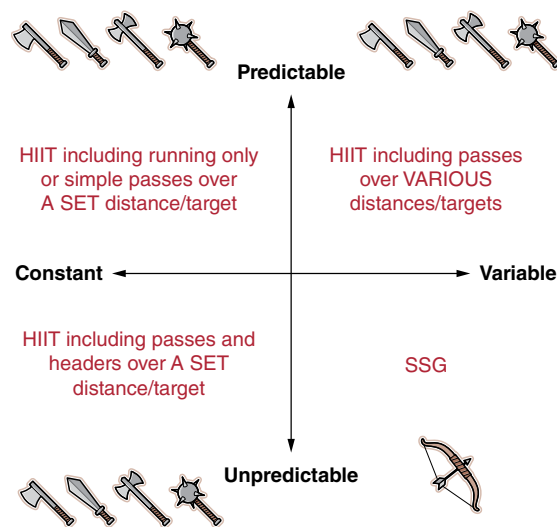


Figure 5.32 The Stratton model and its use for characterizing sport activities with respect to skill acquisition and motor learning, based on the respective levels of variability and randomness of training situations. The five weapons are positioned with respect to the two axes. SSG examples in red.

Adapted from G. Stratton, T. Reilly, A.M. Williams, and D. Richardson, *Youth Soccer: From Science to Performance* (Abingdon, Oxon: Routledge, 2004).

number of both gas analyzers and player availability) and complexity (e.g., disruption of technical skills, potential harm caused by analyzer in case of contact, etc.) of such measures during game play. The $\dot{V}O_2$ demands of run or cycling-based HIIT discussed in the first sections of the chapter are generally examined by testing one athlete after the next consecutively, using a single gas analyzer (one athlete per testing session, 12 athletes tested = 12 experimental sessions, a task that is feasible for any research team). In contrast, when it comes to monitoring the $\dot{V}O_2$ responses of SSGs, 3 to 5 gas analyzers are needed to characterize the $\dot{V}O_2$ demands of multiple players at the same time—a costly endeavor for any organization. In fact, due to the respective interactions between the 6 to 8 players typically involved during GBHIIT, every experimental GBHIIT must involve almost the entire participant pool (with a portion of players using a gas analyzer). This means that if we were to use only one gas analyzer, and considering the intense demands of GBHIIT that preclude repeating a high number of sets on the same day (i.e., player fatigue), and considering that at least two trials per athlete are required to compensate for the well-known SSG-to-SSG variability in locomotor patterns (see chapter 2; in turn, leading to more variability of energetic demands), participants would in effect be required to visit the lab a ridiculous number of times to allow each individual player to be tested. If you add to this the manipulation of some of the programming variables (e.g., exercise intensity, exercise length, and tactical rules, figure 5.31), it becomes almost impossible to monitor the $\dot{V}O_2$ responses of different variations in GBHIIT formats. In the real world, this simply

wouldn't happen. For this reason, the majority of researchers to date have used HR responses to SSGs to characterize the metabolic responses and effects of various programming manipulations, since HR can easily and effectively be collected simultaneously in all players. The aforementioned intermittent exercise HR/ $\dot{V}O_2$ dissociation that naturally arises during the phases that characterize GBHIIT (especially at exercise onset and in recovery, see figures 2.3 and 5.33) adds a degree of uncertainty when using HR as a proxy marker of the oxidative response. As with using blood lactate measures to infer anaerobic energy demands, we must accept the HR/ $\dot{V}O_2$ limitations so as to begin to understand how SSGs can be manipulated to solve the puzzle. While the $\dot{V}O_2$ /HR relationship issue limits direct interpolation of actual $\dot{V}O_2$ demands per se, this issue plays less of a role when assessing the relative effect of GBHIIT variable manipulation.

The limited literature available describing actual $\dot{V}O_2$ responses shows that some forms of SSGs can be used to achieve high levels of $\dot{V}O_2$ (4v4 handball SSGs: >90% $\dot{V}O_{2\max}$ (56) or 2v2 in basketball: 80% $\dot{V}O_{2\max}$ (67)), but not others involving more players (5v5 indoor soccer: peak values reached = 75% $\dot{V}O_{2\max}$ (65)). When it comes to assessing the effects of SSG manipulation on $\dot{V}O_2$ responses in the same population, there is even less data available. The study from Castagna et al. (67) in junior basketball players ($\dot{V}O_{2\max}$ 55 mL·min⁻¹·kg⁻¹) showed that mean $\dot{V}O_2$ responses increased linearly by 5% when player number was reduced from 5v5 (average response 69% of $\dot{V}O_{2\max}$) to 3v3 (74%) and 2v2 (79%). Further considerations regarding the effects of SSG manipulation that follow are based on HR measures only.

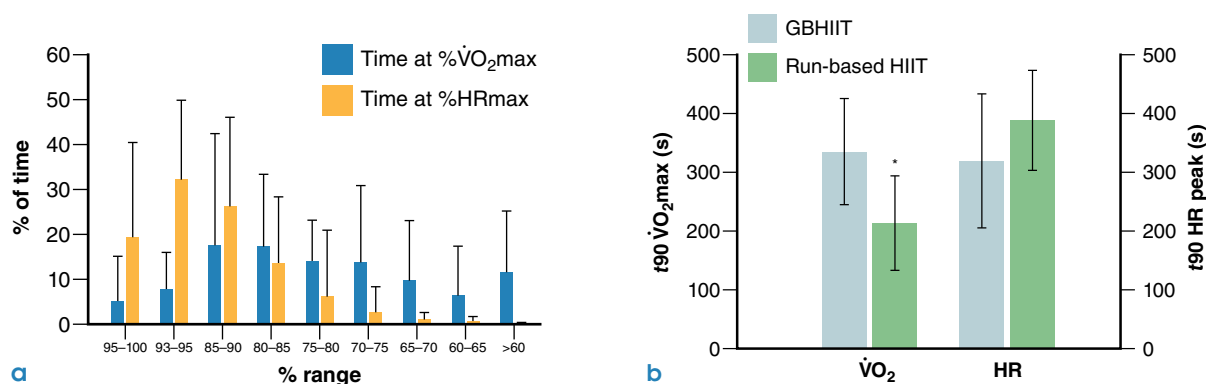


Figure 5.33 (a) Percentage of game time spent in selected $\dot{V}O_2$ and HR intensity zones (mean and 95% CI) during futsal game play (66). (b) Mean (95% CI) time spent above 90% of maximal oxygen uptake (t90 $\dot{V}O_{2\max}$) and above 90% of maximal heart rate (t90 HR peak) during handball SSGs (game-based HIIT, GBHIIT), and HIIT with short intervals (run-based HIIT) (56). *Significant difference between GBHIIT and run-based HIIT.

Work Intensity

The different options available for practitioners to modulate exercise intensity during GBHIIT are summarized in figure 5.31. Varying a player's relative area via changes in either player number and/or field size are the two most adjusted variables used in the field. Since comparable player areas can be achieved using different variations of either player number or field size, HR responses to player area variations have presented mixed findings in the literature. When player number is constant, an increase in field size is generally associated with increased HR responses (since greater space allows for more running, figure 5.34), while in some circumstances findings are not as clear (figure 27.8). The lack of consistency in this regard is likely related to the fact that over very small playing areas, changes in velocity, direction, and contacts tend to increase compared to the larger pitch size (73), and in turn compensate for the reduced running demands (56, 73).

Similarly, lower player numbers in SSGs tend to increase the HR response, due in part to increased ball involvement associated with the reduced team-mate number (figure 5.34). The third most-used

component when it comes to modulating the exercise intensity of SSGs is adjustment of the technical rules, such as varying the number of ball contacts allowed (the lower the number, the greater the HR demands (75)), the defensive style (e.g., man marking, which also increases HR (141)), and adding extra rules (e.g., the need for the attacking team to have all players in a specific area of the pitch—half or third side of the opponent's goal—which tends to increase HR (111)). The inclusion of goalkeepers (GK) into SSGs has shown varying effects on HR (increases without GK when players run more to score on smaller goals, but additionally when GK increase player motivation (112)). Coach encouragement can also motivate player exertion and HR (figure 5.34) (112).

Work Duration

As with long-interval HIIT, the optimal work duration for SSGs of low player number (i.e., 3v3 or 4v4) is between 2 and 4 min (112, 124). The duration should be long enough for $\dot{V}O_2$ to reach high levels (T to $\dot{V}O_{2\max}$) but not so long as to cause excessive fatigue that could impair technical efficiency and the ability to maintain a high T at $\dot{V}O_{2\max}$. In fact,

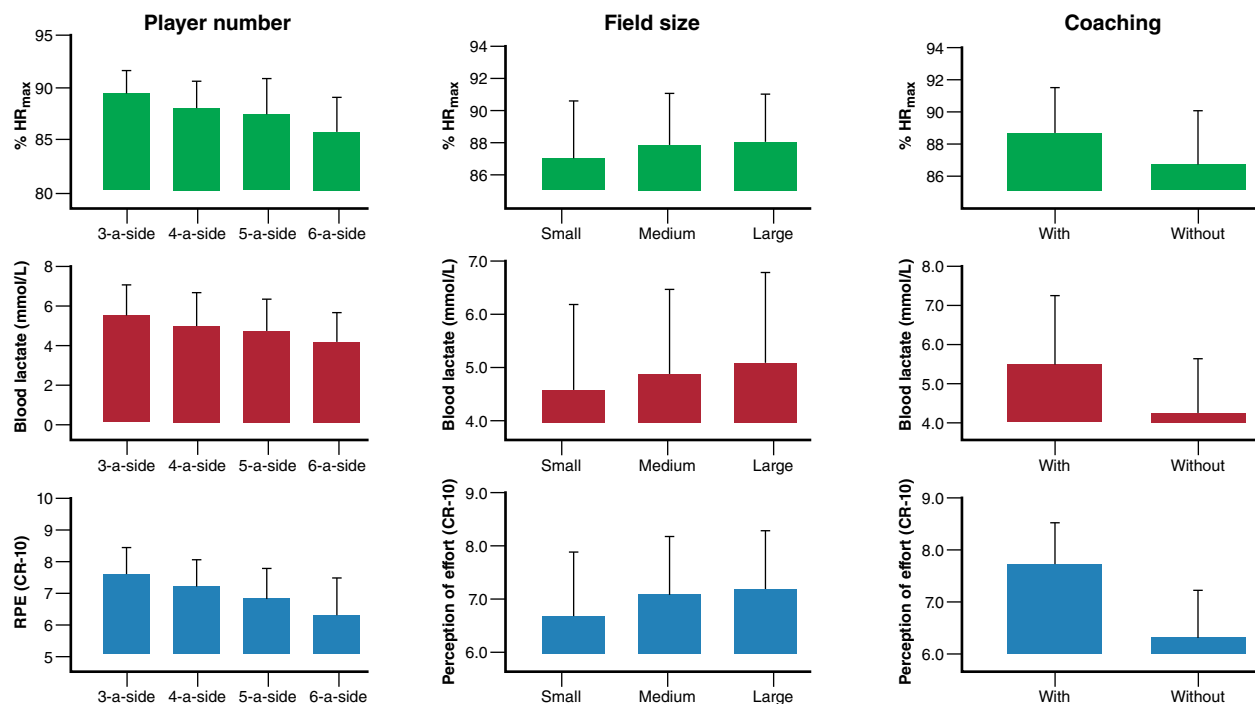


Figure 5.34 Effect of variations in player number, field size, and coach encouragement on heart rate (HR), blood lactate, and rating of perceived exertion (RPE) responses during typical small-sided games (SSGs) in soccer players.

Data from E. Rampinini, F.M. Impellizzeri, C. Castagna, G. Abt, K. Chamari, A. Sassi, and S.M. Marcora, "Factors Influencing Physiological Responses to Small-Sided Soccer Games," *Journal of Sports Sciences* 25 no. 6 (2007): 659-666.

fatigued players will more frequently lose the ball, which inappropriately breaks the activity and lowers overall exercise intensity. Exercise intensity needs also to be balanced with the exercise duration, i.e., longer durations for lower intensities and higher intensities for shorter durations. For example, SSG durations that involve more players (e.g., 8v8) are generally programmed over longer durations (8-10 min) (see chapter 30, Soccer (126)). These longer intervals likely target more threshold-like responses ($>80\%$ - 90% HR_{max} (124)) rather than $\dot{V}O_{2\text{max}}$ ($>90\%$ - 95% HR_{max}) (112).

Playing Surface

In the only study to date (30) investigating the effect of playing surface on metabolic responses to SSGs, active young male adults performed $2 \times 5v5$ SSGs for 10 min (separated by 5 min passive recovery) on either sand, artificial turf, or asphalt. Total distance and sprint number were 2585 ± 208 m and 7 ± 2 ; 3726 ± 121 m and 13 ± 2 ; and 3895 ± 39 m and 21 ± 2 , respectively. The results, summarized in figure 5.35, showed that, in opposition to locomotor demands (asphalt $>$ artificial turf $>$ sand), artificial turf tended to elicit the highest HR, blood lactate, and rating of perceived exertion responses, followed next by asphalt and then sand, respectively (small to moderate differences between all conditions). In practice, however, while it might be feasible to program in recreational beach soccer at some stages of the preseason, it would be difficult to rationalize playing on asphalt given the overall increased injury risk that such a session would expose.

Between-GBHIIT Recovery Periods

As detailed in chapter 4, decreasing duration or increasing intensity of the between-set recovery period impairs the restoration of W' , which can be detrimental to technical performance during the following work period (figure 5.36). Since GBHIITs are first about a player's technical interaction, care should be taken to ensure a minimum level of skill proficiency (limiting fatigue), but also the maintenance of exercise intensity and, in turn, metabolic responses and T at $\dot{V}O_{2\text{max}}$. Rest periods ideally should be 2 to 4 min and completely passive (112).

Anaerobic Glycolytic Energy Contribution During GBHIIT

In contrast to run-based HIIT, research indicates that the oxidative and anaerobic contributions tend to follow the same pattern of response to HIIT variable manipulation during GBHIIT (71). This implies that the general effects of SSG manipulation on anaerobic energy contribution may be similar to those discussed earlier for HR (and hopefully $\dot{V}O_2$) responses. Increases in field dimension, decreases in player number, lowered number of ball contacts, man-to-man marking, goalkeeper integration, and coach encouragement all tend to increase anaerobic contribution to exercise (figures 5.34 and 5.36). In contrast, shorter SSG duration and longer between-SSG breaks tend to lower anaerobic contribution.

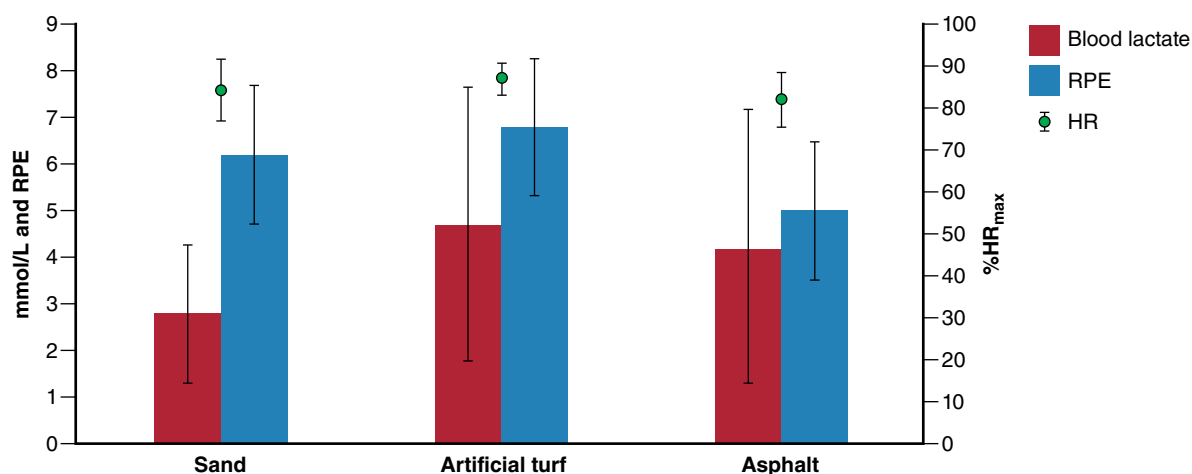


Figure 5.35 Heart rate (HR), blood lactate, and rating of perceived exertion (RPE) response to 5v5 SSGs performed on different surfaces.

Data from J. Brito, P. Krstrup, and A. Rebelo, "The Influence of the Playing Surface on the Exercise Intensity of Small-Sided Recreational Soccer Games," *Human Movement Science* 31 no. 4 (2012): 946-956.

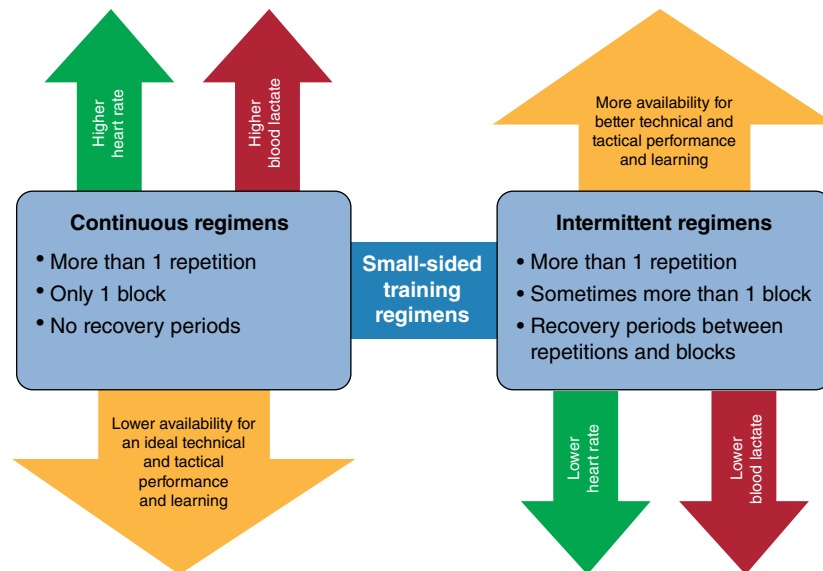


Figure 5.36 Schematic representation of the effect of continuous versus intermittent regimens (and by extension the impact of the between-set recovery characteristics) when using SSGs (182).

Neuromuscular Responses to GBHIIT

The neuromuscular impact of GBHIIT has probably received the least attention in the literature. The work intensity, duration, between-set recovery, and playing surface should all be considered for this aspect of the weapon's target (type 2 and 4 responses).

Work Intensity, Duration, and Between-Set Recovery

Since the actual neuromuscular load is likely to be exercise intensity dependent (the greater the exercise intensity, the more associated mechanical load), the same relationship as described for the oxidative and anaerobic contribution should apply. It is, however, worth noting that the type of neuromuscular fatigue is likely different for large- and small-size SSGs, since it involves different types of locomotion. In fact, while small fields tend to increase mechanical work (i.e., the number and magnitude of acceleration/deceleration and changes of direction, which stress lower-limb extensors such as quads and glutes), larger fields likely create more total distance run (figure 27.8) at higher speeds, which stresses hamstring muscles (157) (see also chapter 30, Soccer, for discussion on the relative contribution of the lower-limb muscle as a function of the type of locomotor actions). In support of the greater neuromuscular load associated with smaller player number SSGs, sprint ability over

5 and 15 m tests (acceleration phases, involving the extensors more) decreased moderately after a 4v4 + GK, but not after an 8v8 + GK (157).

Playing Surface

Playing surface also impacts the development of acute neuromuscular fatigue with GBHIIT independent of other physiological responses. For example, when active young male adults performed two 5v5 SSGs for 10 min (5 min passive recovery between) on either sand, artificial turf, or asphalt, sprint performance wasn't substantially affected. In contrast, there were small reductions in CMJ for artificial turf and asphalt (figure 5.37) (30). As discussed earlier, however, there may be only a limited number of practitioners who would actually program SSGs on asphalt for the reasons mentioned earlier (greater injury risk, etc.). Therefore further research should focus on artificial turf versus natural grass.

The main effects of GBHIIT variable manipulations are summarized in figure 5.38.

Analysis and Comparison of HIIT Weapons

We finish chapter 5 with a short section comparing all formats against one another, in terms of their oxidative, anaerobic glycolytic, and neuromuscular acute responses.

Comparison of $\dot{V}O_2$ Responses Among HIIT Weapons

Throughout chapter 5, we have highlighted the $\dot{V}O_2$ responses to various forms of HIIT. It appears that most HIIT formats, when properly manipulated, can enable athletes to reach $\dot{V}O_{2\max}$. However, important between-athlete and between-HIIT format dif-

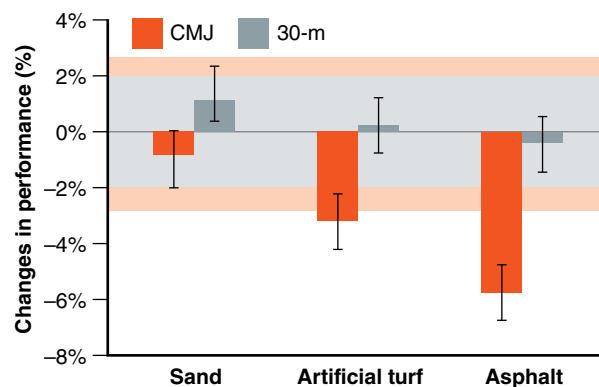


Figure 5.37 Changes in countermovement jump (CMJ) and sprint times following 5v5 SSGs performed on sand, artificial turf, or asphalt. Gray and orange zones represent trivial changes for sprint and CMJ performance, respectively.

Data from J. Brito, P. Krstrup, and A. Rebelo, "The Influence of the Playing Surface on the Exercise Intensity of Small-Sided Recreational Soccer Games," *Human Movement Science* 31 no. 4 (2012): 946-956.

ferences exist with respect to T at $\dot{V}O_{2\max}$. RSS and SIT sessions allow for a limited T at $\dot{V}O_{2\max}$ compared to HIIT that involve long and short intervals (figure 5.39). Combined, data from high-level athletes (90, 136) suggests that long intervals and/or short intervals with a work:relief ratio >1 should enable a greater T at $\dot{V}O_{2\max}$ /exercise time ratio during HIIT sessions. Table 5.2 also provides overview comparisons of the impact of the various HIIT weapons in terms of their physiological response types.

Comparison of the Anaerobic Glycolytic Responses Between HIIT Weapons

The anaerobic glycolytic energy contribution during an HIIT session is HIIT parameter dependent. SIT- and RSS-type HIIT weapons are typically associated with elevated rates of blood lactate accumulation (figure 5.40). During RSS sessions, sprint durations greater than 4 s with work relief intervals less than 20 s lead to the highest blood lactate accumulation. In contrast, SIT sessions require relief interval durations equal to or longer than 4 min to maximize the anaerobic glycolytic energy response. As shown, the manipulation of HIIT variables with short intervals may allow practitioners to vary the level of anaerobic glycolytic energy contribution to a given session. In general, longer interval durations and higher interval intensities (when > $v\dot{V}O_{2\max}$) will elicit

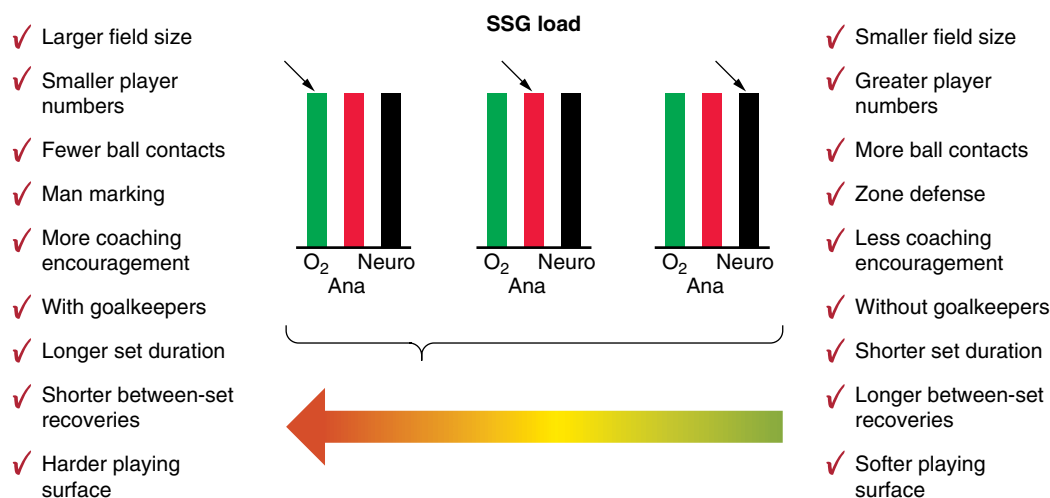


Figure 5.38 Main physiological responses in relation to variations in some of the programming aspects of GBHIIT and small-sided games (SSGs). Note that compared with run-based HIIT, and with the exception of the impact of playing surface, there is a tendency for all physiological responses (oxidative, anaerobic, and neuromuscular systems) to respond in a similar fashion.

Table 5.2 Recommendations for the Design of Run-Based HIIT Protocols for Optimizing Time at Maximal Oxygen Uptake (T at $\dot{V}O_{2\max}$), Ordered From Lowest to Highest in Terms of Expected Oxidative Response

Format	W duration	W intensity	Modality	R duration	R intensity	Reps and series ^(c)	Between-set recovery		Expected T at $\dot{V}O_{2\max}$	Acute demands
							Duration	Intensity		
SIT	> 20 s	All-out	Sport specific	≥ 2 min	Passive	6-10			0-1 min	Peripheral +++
RST	> 4 s (> 30 m or 2 × 15 m)	All-out	COD Jumps Explosive efforts	< 20 s	≈ 55% $\dot{V}O_{2\max}$ /40% V_{IFT}	2-3 RSS (each > 6 sprints)	≥ 6 min	≤ 60-70% $\dot{V}O_{2\max}^{(b)}$	0-3 min	Central + Peripheral +++
Game-based training	> 2-3 min	Self-selected RPE > 7	Sport specific ^(d)	≤ 2 min	Passive	6-10 × 2 min 5-8 × 3 min 4-6 × 4 min			> 8 min	Central ++ Peripheral +++
HIIT with long intervals	2-3 min ^(a)	≥ 95% $\dot{V}O_{2\max}$	Sport specific	≤ 2 min ≥ 4-5 min	Passive ≤ 60-70% $\dot{V}O_{2\max}^{(b)}$	6-10 × 2 min 5-8 × 3 min 4-6 × 4 min			> 10 min	Central +++ Peripheral ++
HIIT with short intervals	≥ 15 s ^(a,c)	100-120% $\dot{V}O_{2\max}$ (85-105% V_{IFT})	Sport specific	< 15 s ≥ 15 s	Passive ≤ 60-70% $\dot{V}O_{2\max}$ (45-55% V_{IFT})	2-3 × ≥ 8-min series	≥ 4-5 min	≤ 60-70% $\dot{V}O_{2\max}^{(b)}$	> 10 min	Central +++ Peripheral ++

Specific recommendation for manipulating various forms of high-intensity interval training (HIIT), including long or short intervals, repeated-sprint training (RST), and sprint interval training (SIT). Intensities are provided as percentages of the speed associated with maximal oxygen uptake ($\dot{V}O_{2\max}$) and the speed reached during the 30-15 intermittent fitness test (V_{IFT}).

(a) To be modulated with respect to exercise mode (longer for cycling vs. running, for example), age, and fitness status (shorter for younger and/or more-trained athletes).

(b) These can also be game-based (moderate intensity) in team sports.

(c) To be modulated with respect to the sport, i.e., longer for endurance and highly trained athletes than team-sport and less-trained athletes.

(d) To be modulated with respect to physiological training objectives (manipulating playing number, pitch area, etc.) and so specific rules are added for the fittest players to compensate for the fitness-related responses, which will parallel the HIIT sessions.

W: work; R: relief; reps: repetitions; RPE: rating of perceived exertion; COD: changes of direction; SSG: small-sided games; T at $\dot{V}O_{2\max}$: time at $\dot{V}O_{2\max}$.

The number of symbols (+) indicate the magnitude of the expected adaptations.

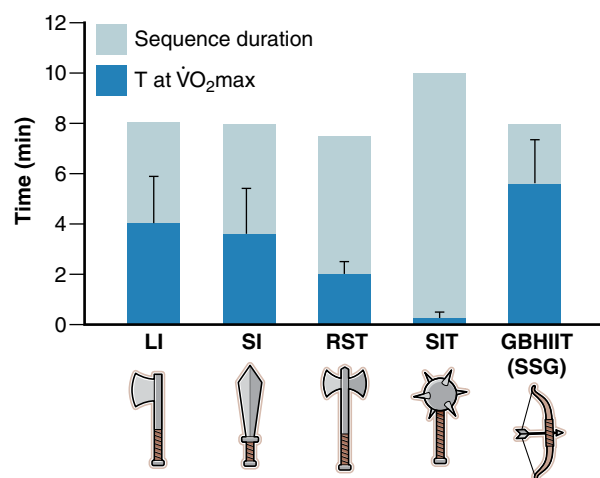


Figure 5.39 Total exercise time and time spent at or near maximal oxygen uptake (T at $\dot{V}O_{2max}$) for typical HIIT formats: 2×2 min/1 min long intervals (LI), 1×8 min series of short intervals (15 s/15 s, SI), a repeated sprint sequence (RST, 2 sets of 6 sprints), a sprint interval session (SIT, 4 sprint repetitions), and an 8 min small-sided game (SSG, one of the most common forms of game-based HIIT, GBHIIT).

Data from M. Buchheit and P.B. Laursen, "High-Intensity Interval Training, Solutions to the Programming Puzzle: Part I: Cardiopulmonary Emphasis," *Sports Medicine* 43, no. 5 (2013): 313-338.

higher blood lactate levels. While the evidence is limited, increasing the intensity of exercise during the recovery interval in short HIIT bouts may also increase blood lactate accumulation. Interestingly, anaerobic glycolytic energy release can also be manipulated during GBHIIT (i.e., SSG) via changes in rules and/or player number and pitch dimension (112) so that blood lactate accumulation can be maintained at low levels despite a prolonged T at $\dot{V}O_{2max}$ (56). A comparison of the anaerobic glycolytic energy contribution and response type to different HIIT weapons is shown in table 5.3 and figure 5.40.

Comparison of the Neuromuscular Responses Among HIIT Weapons

While the magnitude of neuromuscular loading during HIIT can be modulated through the manipulation of HIIT variables (e.g., work intensity or duration, exercise mode or pattern; figure 5.41), the responses are highly athlete profile dependent, with endurance-type athletes showing low levels of acute fatigue and speed decrement, and team sport athletes typically showing high levels of neuromuscular fatigue following HIIT (figure 5.42). There is likely a

Table 5.3 Recommendations for the Design of Run-Based HIIT Protocols With Respect to Blood Lactate Accumulation, Ordered From Lowest to Highest in Terms of Expected Lactate Response

Format	W duration	W intensity	Modality	R duration	R intensity	Expected initial rate of blood lactate accumulation
HIIT with short intervals	≥ 20 s <30s	$<100\% \dot{V}O_{2max}$ ($<89\% V_{IFT}$)	Straight line	≥ 20 s <30s	$\approx 55\% \dot{V}O_{2max}$ ($40\% V_{IFT}$)	$<5 \text{ mmol} \cdot \text{L}^{-1} \cdot 5 \text{ min}^{-1}$
	<15 s	$<120\% \dot{V}O_{2max}$ ($<100\% V_{IFT}$)	Straight line	≥ 20 s	Passive	$<5 \text{ mmol} \cdot \text{L}^{-1} \cdot 5 \text{ min}^{-1}$
Game-based training	3-4 min	Self-selected RPE >7	Sport specific	≤ 2 min ≥ 4 -5 min	Passive $55\% \dot{V}O_{2max}$ ($40\% V_{IFT}$)	$\leq 5 \text{ mmol} \cdot \text{L}^{-1} \cdot 5 \text{ min}^{-1}$
HIIT with long intervals	<2 min	$<100\% \dot{V}O_{2max}$	Straight line	2 min	Passive	$\approx 5 \text{ mmol} \cdot \text{L}^{-1} \cdot 5 \text{ min}^{-1}$
HIIT with short intervals	>25 s	$>110\% \dot{V}O_{2max}$ ($>90\% V_{IFT}$)	COD	>15 s <30 s	$60\%-70\% \dot{V}O_{2max}$ ($45\%-55\% V_{IFT}$)	$\approx 6\text{-}7 \text{ mmol} \cdot \text{L}^{-1} \cdot 5 \text{ min}^{-1}$
HIIT with long intervals	>3 min	$\geq 95\% \dot{V}O_{2max}$	Straight line, sand, hills	>3 min	Passive	$\approx 6\text{-}7 \text{ mmol} \cdot \text{L}^{-1} \cdot 5 \text{ min}^{-1}$
RST	<3 s	All-out	$45^\circ\text{-}90^\circ$ COD	>20 s	Passive	$\leq 10 \text{ mmol} \cdot \text{L}^{-1} \cdot 5 \text{ min}^{-1}$
RST	>4 s	All-out	Straight line + jump	<20 s	$\approx 55\% \dot{V}O_{2max}$ ($40\% V_{IFT}$)	$>10 \text{ mmol} \cdot \text{L}^{-1} \cdot 5 \text{ min}^{-1}$
SIT	>20 s	All-out	Straight line	>120 s	Passive	$>10 \text{ mmol} \cdot \text{L}^{-1} \cdot 5 \text{ min}^{-1}$

Specific recommendation for manipulating various forms of high-intensity interval training (HIIT), including long or short intervals, repeated-sprint training (RST), and sprint interval training (SIT). Intensities are provided as percentages of the speed associated with maximal oxygen uptake ($\dot{V}O_{2max}$) and the speed reached in the 30-15 intermittent fitness test (V_{IFT}). W: work; R: relief; RPE: rating of perceived exertion; COD: changes of direction.

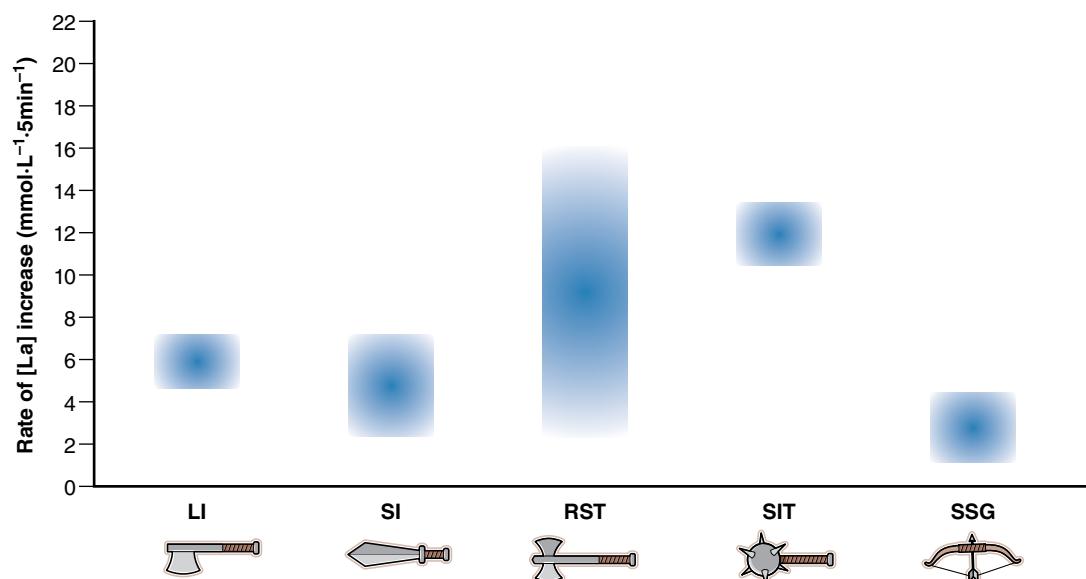


Figure 5.40 Range of lactate accumulation typically observed for the five different HIIT formats.

Data from M. Buchheit and P.B. Laursen, "High-Intensity Interval Training, Solutions to the Programming Puzzle: Part II: Anaerobic Energy, Neuromuscular Load and Practical Applications," *Sports Medicine* 43 no. 10 (2013): 927-954.

bell-shaped relationship between exercise intensity and acute neuromuscular performance responses, with too low ($\leq 85\%$ $\dot{V}O_{2\max}$) and too high (all-out) an intensity having not enough and acute detrimental effects, respectively (figure 5.43). Work intensities $>80\%$ - 85% $\dot{V}O_{2\max}$ require recruitment of fast-twitch fibers (97), induce post-activation potentiation, and possibly lead to long-term structural adaptations that allow fatigue resistance to high-speed running (161). In contrast, supramaximal-to-maximal ($\geq 120\%$ $\dot{V}O_{2\max}$, $\geq 100\%$ V_{IFT}) intensity exercises are likely associated with acute impairments in muscular performance. The residual fatigue from HIIT sessions that persists over time may have large implications for subsequent training (carry-over effect), but there is limited data documenting the recovery course of neuromuscular function following HIIT. Finally, in addition to mechanical work intensity, the associated metabolic responses and potentially accumulated metabolites within muscle should also be taken into account when examining the acute neuromuscular load of a given ses-

sion. Coaches can choose to balance the level of neuromuscular engagement associated with a given HIIT format, based on both the expected training-induced adaptations (either through the HIIT session itself or the associated sessions and possible additive effects) and the acute changes in neuromuscular performance (table 5.4). Running pattern (e.g., COD, introduction of jumps during the recovery periods), exercise mode (e.g., cycling, running, bouncing), or ground surfaces (e.g., pavement, synthetic track, grass, sand, treadmill) and terrain (uphill, downhill) also may have direct implications on traumatic and overuse injury risk, and should be chosen for programming based on a risk-benefit approach (figure 5.43). Similarly, before programming an HIIT training cycle, coaches should also consider that in team sports, physical fitness is unlikely to have the same impact on match running performance for all players, as playing position, systems of play, and individual playing styles directly affect the relationship between physical fitness and match running performance (59, 61, 130, 139).

Table 5.4 Recommendations for the Design of Run-Based HIIT Protocols in Reference to Acute Neuromuscular Performance and Potential Injury Risk

Format	W duration	W intensity	Modality	Ground surface	R duration	R intensity	Acute change in muscular performance ^(a)	Injury risk level
Game-based training	> 2-3 min	Self-selected RPE > 7	Sport specific	Sport specific	≤ 2 min	Passive 55% $\dot{V}O_2$ max (40% V_{IFT})	SSG format-dependent	Traumatic ++ (contacts, joint sprain) Overuse +
HIIT with long intervals	> 2-3 min ^(b)	≥ 95% $\dot{V}O_2$ max	Straight line	Grass or treadmill	2 min	Passive	From improved ++ to impaired ++	Traumatic - Overuse +
	> 2-3 min ^(b)	≥ 95% $\dot{V}O_2$ max	Straight line	Track	4-5 min	60-70% $\dot{V}O_2$ max (45-55% V_{IFT})	From improved + to impaired +	Traumatic ++ (tendons) Overuse +++
	> 2-3 min ^(b)	≥ 85% $\dot{V}O_2$ max	Hill	Road	2 min	Passive	From improved + to impaired +	Traumatic - Overuse ++ (downhill = shocks)
HIIT with short intervals	< 15 s	< 120% $\dot{V}O_2$ max (< 100% V_{IFT})	Straight line	Track, indoor	> 15 s but < 30 s	60-70% $\dot{V}O_2$ max (45-55% V_{IFT})	From improved + to impaired +	Traumatic - Overuse - (since short series)
	≤ 20 s	< 110% $\dot{V}O_2$ max (> 90% V_{IFT})	Straight line	Track, indoor	≤ 20 s	Passive	From improved + to impaired ++	Traumatic - Overuse ++ (since long series)

	≤ 20 s	< 110% $\dot{V}O_{2\max}$ (> 90% V_{IFT})	COD	Track, indoor	≤ 20 s	Passive	From improved + to impaired ++	Traumatic ++ (ankle sprain) Overuse ++
	≤ 20 s	< 110% $\dot{V}O_{2\max}$ (> 90% V_{IFT})	COD	Grass	≤ 20 s	Passive	From improved - to impaired +++	Traumatic ++ (ankle sprain and adductors) Overuse +
RST	≤ 5 s	All-out	Distance < 20 m 45-90 degree COD	Sport specific	≤ 25 s	passive	Impaired - to ++	Traumatic ++ (ankle sprain)
	> 3 s	All-out	Straight line > 20 m	Sport specific	≤ 30 s	≈ 55% $\dot{V}O_{2\max}$ / 40% V_{IFT}	Impaired + to +++	Traumatic ++ (hamstring)
	≤ 3 s	All-out	Straight line < 20 m, COD, and jump	Sport specific	≤ 30 s	≈ 55% $\dot{V}O_{2\max}$ / 40% V_{IFT}	Impaired ++ to ++++	Traumatic -
SIT	> 20 s	All-out	Straight line	Sport specific	> 120 s	Passive	Impaired ++ to ++++	Traumatic ++++ (hamstring)

Specific recommendations for manipulating various forms of high-intensity interval training (HIIT) sessions including long or short intervals, repeated-sprint training (RST), and sprint interval training (SIT). Intensities are provided as percentages of the speed associated with maximal oxygen uptake ($\dot{V}O_{2\max}$) and the final speed reached in the 30-15 intermittent fitness test (V_{IFT}). Comparisons between muscular fatigue following runs over different surfaces are adapted from the work of Sassi et al. (A. Sassi, A. Stefanescu, P. Menaspá, A. Bosio, M. Riggio, and E. Rampinini, "The Cost of Running on Natural Grass and Artificial Turf Surfaces," *J Strength Cond Res.* 25, no. 3 (2011): 606-611.) (with an energy cost of running lower for hard surface < sand < grass < treadmill) and Gains et al. (G.L. Gains, A.N. Swedenhjelm, J.L. Mayhew, H.M. Bird, and J.J. Houser, "Comparison of Speed and Agility Performance of College Football Players on Field Turf and Natural Grass," *J Strength Cond Res.* 24, no. 10 (2010): 2613-2617). While additional combinations between the different surfaces and exercise modes can be implemented (e.g., uphill running on a treadmill), these examples illustrate the main logic behind the selection of HIIT variables. The level of injury rate is estimated based on the combination of running speed, total distance at high intensity, ground surface, and specific running patterns and is expressed as a function of the number of (+) symbols.

(a) Fatigue responses are likely athlete-dependent.

(b) To be modulated with respect to exercise mode (longer for cycling vs. running for example), age, and fitness status (shorter for younger and/or more trained athletes).

SSG: small-sided game; W: work; R: relief; RPE: rating of perceived exertion.

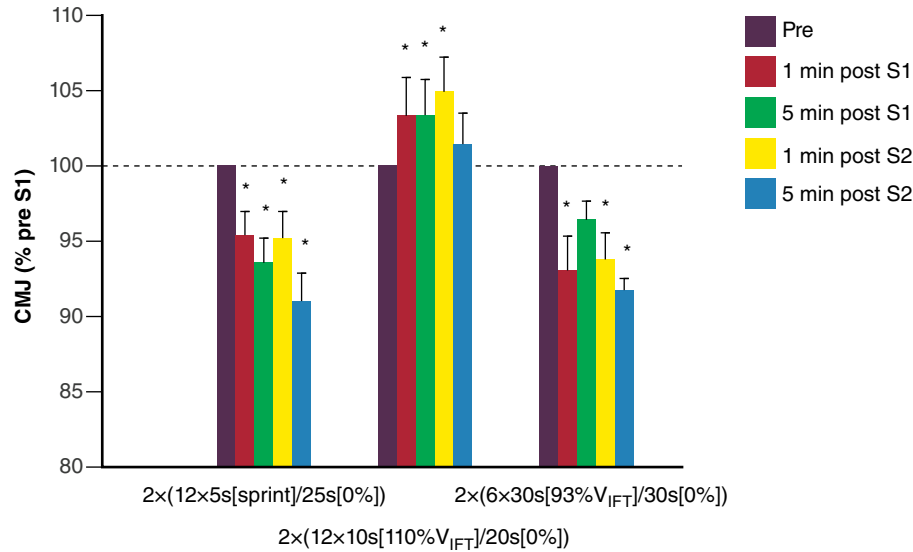


Figure 5.41 Changes in countermovement jump (CMJ) height following three different high-intensity interval training sessions (repeated-sprint sequence and high-intensity interval training of either 10 s/20 s or 30 s/30 s format) performed over two series (146). S1: first series; S2: second series; V_{IFT} : peak speed reached in the 30-15 intermittent fitness test; * indicates moderate standardized difference; # indicates large standardized difference.

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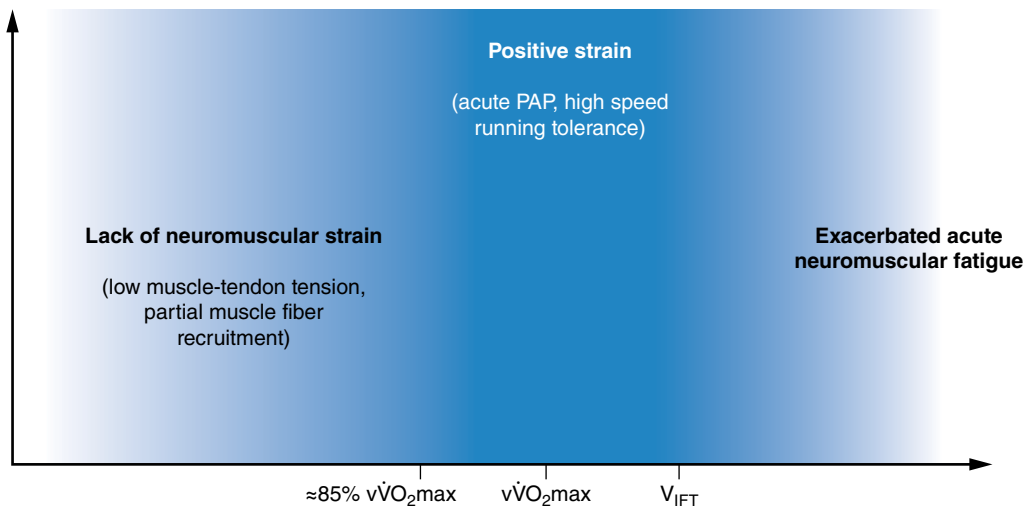


Figure 5.42 Neuromuscular load and associated performance outcomes as a function of running speed during typical high-intensity interval training sessions. PAP: post-activation potentiation; V_{IFT} : peak speed reached at the end of the 30-15 intermittent fitness test; vVO_{2max} : lowest running speed required to elicit maximal oxygen uptake.

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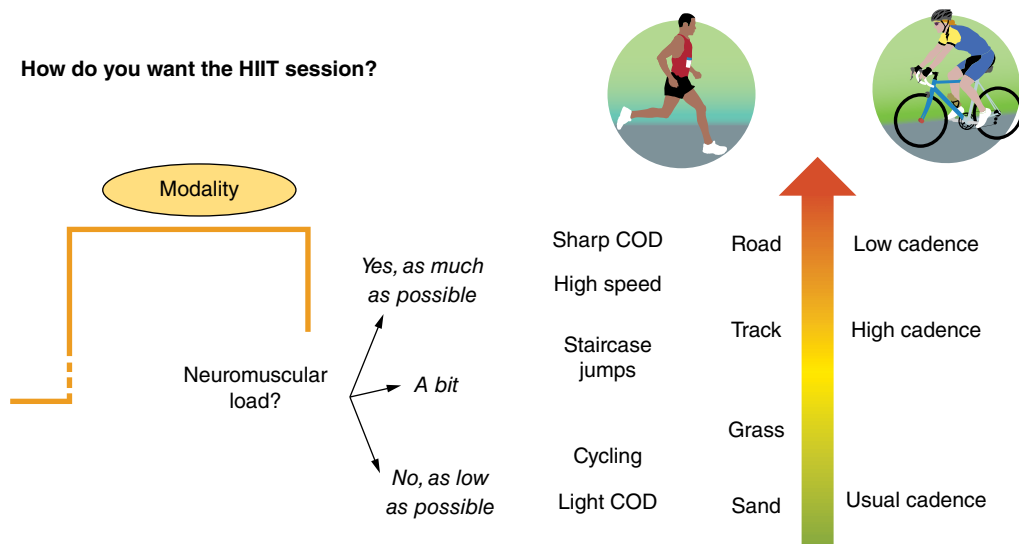


Figure 5.43 The effect of work modality during HIIT sessions on the expected neuromuscular and musculoskeletal demands.

Summary

To summarize chapter 5, we have gone deep into the science to review how the key moving parts of HIIT variable manipulation (chapter 4) can be used to adjust the physiological responses in the five key HIIT weapons to hit our desired targets. As illustrated originally in figure 1.5, long intervals (figure 4.13) can effectively be manipulated to hit type 3 and 4 targets; short intervals (figure 4.14) are versatile formats that can be used to hit the majority of HIIT targets (types 1-4); RST (figure 4.15) is useful for hitting type 4 and 5 targets; while SIT

(figure 4.16) is a specialist weapon used to hit type 5 targets. The master team and racket sport weapon is the GBHIIT (figure 4.17), which is a highly flexible tool for hitting type 2 through 4 targets. The decision tree model originally offered in figure 1.5 provides a roadmap solution to the programming puzzle, which can be used alongside the fine-tuning details offered in this chapter to help more precisely guide the manipulation of HIIT sessions in line with target-type goals. The complexities and the science encompassing the integration of concurrent type 6 speed and strength targets, shown in figure 1.5, are described in chapter 6.

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